Lecture Notes: Tensor Calculus

Based on Youtube series by eigenchris

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1 Multi-variable Calculus Review

For this lecture notes on Tensor Calculus, there are some concepts and notes that we need to quickly review on multi-variable calculus, which need to be clear and understood before proceeding.
Here we'll give a brief review of the following topics:

- Single-variable Derivatives and chain rule;
- Single-variable Integrals and differentials (change of variable);
- Multi-variable Partial derivatives and chain rule;
- Gradient of a function;
- Directional Derivatives
- Multi-variable Integrals and differentials;
- Arc length, velocity vector tangent to a curve.

1.1 Single-Variable Derivatives and Chain Rule

The derivative of a function $f(x)$ with respect to x measures the instantaneous rate of change:

$$\frac{\mathrm{d}f}{\mathrm{d}x} = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

Basic derivative rules:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}(x^n) &= nx^{n-1} \\
\frac{\mathrm{d}}{\mathrm{d}x}(e^x) &= e^x \\
\frac{\mathrm{d}}{\mathrm{d}x}(\sin x) &= \cos x \\
\frac{\mathrm{d}}{\mathrm{d}x}(\cos x) &= -\sin x \\
\frac{\mathrm{d}}{\mathrm{d}x}(\ln x) &= \frac{1}{x}
\end{aligned}$$

Sum and product rules:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}(f+g) &= \frac{\mathrm{d}f}{\mathrm{d}x} + \frac{\mathrm{d}g}{\mathrm{d}x} \\
\frac{\mathrm{d}}{\mathrm{d}x}(fg) &= f\frac{\mathrm{d}g}{\mathrm{d}x} + g\frac{\mathrm{d}f}{\mathrm{d}x}
\end{aligned}$$

Chain rule: For a composition of functions $f(g(x))$:

$$\frac{\mathrm{d}}{\mathrm{d}x} f(g(x)) = \frac{\mathrm{d}f}{\mathrm{d}g} \cdot \frac{\mathrm{d}g}{\mathrm{d}x}$$

More explicitly, if $y = f(u)$ and $u = g(x)$, then:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}u} \cdot \frac{\mathrm{d}u}{\mathrm{d}x}$$

Example: Let $y = \sin(x^2)$. Setting $u = x^2$, we have $y = \sin(u)$:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \cos(u) \cdot 2x = 2x\cos\big(x^2\big)$$

1.2 Single-Variable Integrals and Change of Variable

The definite integral of $f(x)$ from a to b represents the signed area under the curve:

$$\int_a^b f(x)\,dx$$

Fundamental Theorem of Calculus: If $F(x)$ is an antiderivative of $f(x)$ (i.e., $\frac{\mathrm{d}F}{\mathrm{d}x} = f$), then:

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

Basic integrals:

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

$$\int \frac{1}{x}\,dx = \ln|x| + C$$

$$\int e^x\,dx = e^x + C$$

$$\int \sin x\,dx = -\cos x + C$$

$$\int \cos x\,dx = \sin x + C$$

Change of Variable (Substitution): To integrate $f(g(x))g'(x)$, substitute $u = g(x)$, so that $du = g'(x)\,dx$:

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du$$

For definite integrals, the limits change accordingly: if $u = g(x)$, then $x = a \to u = g(a)$ and $x = b \to u = g(b)$.

Example: Evaluate $\int x\cos(x^2)\,dx$. Let $u = x^2$, then $du = 2x\,dx$, so $x\,dx = \frac{1}{2}du$:

$$\int x\cos\big(x^2\big)\,dx = \int \cos(u) \cdot \frac{1}{2}\,du = \frac{1}{2}\sin(u) + C = \frac{1}{2}\sin\big(x^2\big) + C$$

1.3 Multi-Variable Partial Derivatives and Chain Rule

For a function $f(x, y)$ of multiple variables, the **partial derivative** with respect to one variable treats all other variables as constants.

Notation:

$$\frac{\partial f}{\partial x}, \quad \frac{\partial f}{\partial y}, \quad f_x, \quad f_y$$

Definition:

$$\frac{\partial f}{\partial x} = \lim_{h \to 0} \frac{f(x+h, y) - f(x, y)}{h}$$

Example: For $f(x, y) = x^2 y + \sin(xy)$:

$$\frac{\partial f}{\partial x} = 2xy + y\cos(xy)$$
$$\frac{\partial f}{\partial y} = x^2 + x\cos(xy)$$

Multivariable Chain Rule: Consider $z = f(x, y)$ where $x = x(t)$ and $y = y(t)$ are functions of t. Then:

$$\frac{\mathrm{d}z}{\mathrm{d}t} = \frac{\partial f}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial f}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t}$$

More generally, if $z = f(x, y)$ and $x = x(u, v)$, $y = y(u, v)$, then:

$$\frac{\partial z}{\partial u} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial u}$$

Example: Let $z = x^2 + y^2$ with $x = r\cos\theta$ and $y = r\sin\theta$. Then:

$$\begin{aligned}\frac{\partial z}{\partial r} &= \frac{\partial z}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial r} \\ &= (2x)(\cos\theta) + (2y)(\sin\theta) \\ &= 2r\cos^2\theta + 2r\sin^2\theta = 2r\end{aligned}$$

Einstein Summation Convention: Using indexed notation where x^i represents the i-th coordinate, the chain rule can be written compactly as:

$$\frac{\partial f}{\partial u^j} = \frac{\partial f}{\partial x^i}\frac{\partial x^i}{\partial u^j}$$

where summation over the repeated index i is implied (no explicit $\sum$ symbol needed). This convention will be used extensively in tensor calculus.

1.4 Gradient of a Function

The **gradient** of a scalar function $f(x, y, z)$ is a vector field that points in the direction of greatest rate of increase of the function.

Definition: For a function $f : \mathbb{R}^n \to \mathbb{R}$, the gradient is:

$$\boldsymbol{\nabla} f = \nabla f = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{pmatrix}$$

In three dimensions:

$$\boldsymbol{\nabla} f = \frac{\partial f}{\partial x}\hat{i} + \frac{\partial f}{\partial y}\hat{j} + \frac{\partial f}{\partial z}\hat{k}$$

Geometric interpretation:

- $\boldsymbol{\nabla} f$ points in the direction of steepest ascent of f
- $|\boldsymbol{\nabla} f|$ gives the rate of change in that direction
- $\boldsymbol{\nabla} f$ is perpendicular to level curves (2D) or level surfaces (3D) of f

Example: For $f(x, y) = x^2 + y^2$:

$$\boldsymbol{\nabla} f = 2x\hat{i} + 2y\hat{j} = 2(x\hat{i} + y\hat{j})$$

At point $(1, 1)$: $\boldsymbol{\nabla} f = 2\hat{i} + 2\hat{j}$, pointing radially outward.

Applications:

- *Optimization*: Finding maxima/minima (critical points where $\boldsymbol{\nabla} f = 0$)
- *Physics*: Conservative forces $\vec{F} = -\boldsymbol{\nabla} U$ (gravitational, electric fields)
- *Machine learning*: Gradient descent for minimizing loss functions
- *Fluid dynamics*: Pressure gradients drive fluid flow

1.5 Directional Derivatives

While partial derivatives measure the rate of change along coordinate axes, the **directional derivative** measures how fast a function changes in an arbitrary direction.

Definition: The directional derivative of f at point $\vec{a}$ in the direction of unit vector $\vec{u}$ is:

$$D_{\vec{u}} f(\vec{a}) = \lim_{h \to 0} \frac{f(\vec{a} + h\vec{u}) - f(\vec{a})}{h}$$

This gives the instantaneous rate of change of f as we move from $\vec{a}$ in direction $\vec{u}$.

Key requirement: $\vec{u}$ must be a *unit vector* (i.e., $|\vec{u}| = 1$) for the directional derivative to represent the true rate of change per unit distance.

Formula using gradient: The directional derivative can be computed using the dot product:

$$D_{\vec{u}} f = \boldsymbol{\nabla} f \cdot \vec{u} = |\boldsymbol{\nabla} f||\vec{u}| \cos\theta = |\boldsymbol{\nabla} f| \cos\theta$$

where θ is the angle between $\boldsymbol{\nabla} f$ and $\vec{u}$.

Important properties:

- **Maximum rate:** $D_{\vec{u}}f$ is maximized when $\vec{u}$ points in the direction of $\boldsymbol{\nabla} f$ (i.e., $\theta = 0$):

$$\max_{\vec{u}} D_{\vec{u}}f = |\boldsymbol{\nabla} f|$$

- **Minimum rate:** $D_{\vec{u}}f$ is minimized when $\vec{u}$ points opposite to $\boldsymbol{\nabla} f$ (i.e., $\theta = \pi$):

$$\min_{\vec{u}} D_{\vec{u}}f = -|\boldsymbol{\nabla} f|$$

- **Zero rate:** $D_{\vec{u}}f = 0$ when $\vec{u}$ is perpendicular to $\boldsymbol{\nabla} f$ (i.e., $\theta = \pi/2$), meaning along level curves/surfaces

Example 1: Find the directional derivative of $f(x, y) = x^2 + xy$ at $(1, 2)$ in the direction toward $(4, 6)$.

First, compute the gradient:

$$\boldsymbol{\nabla} f = (2x + y)\hat{i} + x\hat{j} \quad \Rightarrow \quad \boldsymbol{\nabla} f(1, 2) = 4\hat{i} + \hat{j}$$

The direction vector from $(1, 2)$ to $(4, 6)$ is $\vec{v} = 3\hat{i} + 4\hat{j}$. Normalize it:

$$\vec{u} = \frac{\vec{v}}{|\vec{v}|} = \frac{3\hat{i} + 4\hat{j}}{\sqrt{9 + 16}} = \frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}$$

The directional derivative is:

$$D_{\vec{u}}f = \boldsymbol{\nabla} f \cdot \vec{u} = (4\hat{i} + \hat{j}) \cdot \left(\frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}\right) = \frac{12}{5} + \frac{4}{5} = \frac{16}{5}$$

Example 2: For $f(x, y) = x^2 + y^2$ at $(3, 4)$, find the direction of maximum increase and the maximum rate.

$$\boldsymbol{\nabla} f = 2x\hat{i} + 2y\hat{j} \quad \Rightarrow \quad \boldsymbol{\nabla} f(3, 4) = 6\hat{i} + 8\hat{j}$$

The direction of maximum increase is along $\boldsymbol{\nabla} f$. The unit vector is:

$$\vec{u}_{\max} = \frac{6\hat{i} + 8\hat{j}}{\sqrt{36 + 64}} = \frac{6\hat{i} + 8\hat{j}}{10} = \frac{3}{5}\hat{i} + \frac{4}{5}\hat{j}$$

The maximum rate of increase is:

$$|\boldsymbol{\nabla} f(3, 4)| = \sqrt{36 + 64} = 10$$

1.6 Multi-Variable Integrals and Differentials

Double and triple integrals extend the concept of integration to functions of multiple variables, computing volumes under surfaces or higher-dimensional analogues.

Double integral: For a function $f(x, y)$ over region R:

$$\iint_R f(x, y)\, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)\, dy\, dx$$

where $dA = dx\,dy$ is the area element.

Triple integral: For a function $f(x, y, z)$ over region V:

$$\iiint_V f(x, y, z)\,dV$$

where $dV = dx\,dy\,dz$ is the volume element.

Change of Variables (Jacobian): When transforming coordinates $(x, y) \to (u, v)$, the area element transforms according to:

$$dx\,dy = |J|\,du\,dv$$

where J is the **Jacobian determinant**:

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}$$

For three dimensions $(x, y, z) \to (u, v, w)$:

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

Example - Polar coordinates: For $x = r\cos\theta$, $y = r\sin\theta$:

$$J = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r\cos^2\theta + r\sin^2\theta = r$$

Therefore: $dx\,dy = r\,dr\,d\theta$

This explains why integrals in polar coordinates have the extra factor of r:

$$\iint_R f(x, y)\,dx\,dy = \iint_{R'} f(r\cos\theta, r\sin\theta)\,r\,dr\,d\theta$$

Differential forms: A differential df of a function $f(x, y)$ is:

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy$$

This represents the infinitesimal change in f due to infinitesimal changes dx and dy in the coordinates. More generally:

$$df = \boldsymbol{\nabla} f \cdot d\vec{r}$$

where $d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$ is the displacement vector.

Einstein notation: Using indexed coordinates x^i, the differential can be written compactly as:

$$df = \frac{\partial f}{\partial x^i}dx^i$$

where summation over repeated index i is implied. This notation emphasizes that the differential is a linear combination of coordinate differentials weighted by partial derivatives.

1.7 Arc Length and Velocity Vector Tangent to a Curve

A **parametric curve** in space is described by a position vector $\vec{r}(t)$ that depends on parameter t:

$$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$$

Velocity vector: The derivative of position with respect to the parameter gives the velocity:

$$\vec{v}(t) = \frac{\mathrm{d}\vec{r}}{\mathrm{d}t} = \frac{\mathrm{d}x}{\mathrm{d}t}\hat{i} + \frac{\mathrm{d}y}{\mathrm{d}t}\hat{j} + \frac{\mathrm{d}z}{\mathrm{d}t}\hat{k} = \dot{x}\hat{i} + \dot{y}\hat{j} + \dot{z}\hat{k}$$

The velocity vector $\vec{v}(t)$ is **tangent to the curve** at each point and points in the direction of motion.

Speed: The magnitude of the velocity vector gives the speed:

$$|\vec{v}(t)| = \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}z}{\mathrm{d}t}\right)^2}$$

Unit tangent vector: Normalizing the velocity gives a unit vector tangent to the curve:

$$\vec{T}(t) = \frac{\vec{v}(t)}{|\vec{v}(t)|}$$

Arc length: The length of a curve from $t = a$ to $t = b$ is:

$$s = \int_a^b |\vec{v}(t)|\, dt = \int_a^b \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}z}{\mathrm{d}t}\right)^2}\, dt$$

In differential form:

$$ds = |\vec{v}|\, dt = \sqrt{dx^2 + dy^2 + dz^2}$$

Arc length parameter: We can reparametrize a curve using arc length s as the parameter. Then:

$$\frac{\mathrm{d}\vec{r}}{\mathrm{d}s} = \vec{T}$$

is automatically a unit tangent vector (since $\left|\frac{\mathrm{d}\vec{r}}{\mathrm{d}s}\right| = 1$).

Example: Consider the helix $\vec{r}(t) = \cos(t)\hat{i} + \sin(t)\hat{j} + t\hat{k}$ for $0 \le t \le 2\pi$.

Velocity vector:

$$\vec{v}(t) = -\sin(t)\hat{i} + \cos(t)\hat{j} + \hat{k}$$

Speed:

$$|\vec{v}(t)| = \sqrt{\sin^2(t) + \cos^2(t) + 1} = \sqrt{2}$$

Arc length from $t = 0$ to $t = 2\pi$:

$$s = \int_0^{2\pi} \sqrt{2}\, dt = 2\pi\sqrt{2}$$

Unit tangent vector:

$$\vec{T}(t) = \frac{1}{\sqrt{2}}\left(-\sin(t)\hat{i} + \cos(t)\hat{j} + \hat{k}\right)$$

Connection to tensor calculus: In general curvilinear coordinates, the tangent vectors to coordinate curves form the basis vectors of the coordinate system. Understanding how tangent vectors transform is fundamental to tensor analysis.

2 Basis vectors as partial derivatives and Jacobian

In two-dimensional Euclidean space, a point is described by Cartesian coordinates

$$(x, y)$$

The position vector is

$$\mathbf{R} = x\vec{e_x} + y\vec{e_y}$$

where the basis vectors are

$$\vec{e_x} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \vec{e_y} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

These satisfy:

$$\vec{e_x} \cdot \vec{e_x} = 1, \quad \vec{e_y} \cdot \vec{e_y} = 1, \quad \vec{e_x} \cdot \vec{e_y} = 0.$$

Thus, $\{\vec{e_x}, \vec{e_y}\}$ is an orthonormal basis.
A key feature:

$$\frac{\partial \vec{e_x}}{\partial x} = \frac{\partial \vec{e_x}}{\partial y} = \frac{\partial \vec{e_y}}{\partial x} = \frac{\partial \vec{e_y}}{\partial y} = 0.$$

The basis vectors are constant everywhere in space.

We now define polar coordinates (r, θ) by the transformation

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

The position vector becomes

$$\mathbf{R} = r\cos\theta\,\vec{e_x} + r\sin\theta\,\vec{e_y}$$

We define the radial unit vector as

$$\hat{e}_r = \cos\theta\,\vec{e_x} + \sin\theta\,\vec{e_y}.$$

This points directly away from the origin.
And then define the angular unit vector as

$$\hat{e}_\theta = -\sin\theta\,\vec{e_x} + \cos\theta\,\vec{e_y}$$

This points in the direction of increasing θ.
These vectors satisfy

$$\hat{e}_r \cdot \hat{e}_r = 1, \quad \hat{e}_\theta \cdot \hat{e}_\theta = 1, \quad \hat{e}_r \cdot \hat{e}_\theta = 0$$

So they form an orthonormal basis at each point, but polar basis vectors depend on θ

Compute derivatives:

$$\frac{\partial \hat{e}_r}{\partial \theta} = -\sin\theta\, \vec{e_x} + \cos\theta\, \vec{e_y} = \hat{e}_\theta$$

$$\frac{\partial \hat{e}_\theta}{\partial \theta} = -\cos\theta\, \vec{e_x} - \sin\theta\, \vec{e_y} = -\hat{e}_r$$

Thus,

$$\frac{\partial \hat{e}_r}{\partial r} = 0, \qquad \frac{\partial \hat{e}_\theta}{\partial r} = 0$$

Key result:

$$\frac{\partial \hat{e}_r}{\partial \theta} = \hat{e}_\theta, \qquad \frac{\partial \hat{e}_\theta}{\partial \theta} = -\hat{e}_r$$

The basis vectors rotate as θ changes.

Now if you think about this, to find out the directions of the basis vectors, what we do is following the directions of the coordinate lines and in the case of polar coordinates, for the $\vec{\tilde{e}_\theta}$ basis vector, since the coordinate lines are actually curved, we need to pick the straightest possible direction when moving along it, which results in being the vector tangent to the curve.

This reasoning gives us a hint for the next important reinterpretation of basis vectors, as represented by a partial derivative of the position vector R

$$\vec{e_x} \equiv \frac{\partial \vec{R}}{\partial x} \tag{1}$$

If you carefully think about it, it might look weird at first, but it will eventually make sense. Think about **R** as a function of the coordinates x and y for example, and then consider adding a small value Δx to the x coordinate. The difference between these two position vectors results in a vector that is parallel to the x coordinate axis. So this resultant vector has the direction of the basis vector $\vec{e_x}$.
This observation is interesting, but the magnitude of this difference vector depends on the value of Δx used.
This can be solved by dividing the resultant vector by the quantity Δx, which turns it into a *rate of change* of the position with respect to the coordinate. Taking the limit as Δx approaches 0:

$$\frac{\partial \vec{R}}{\partial x} = \lim_{\Delta x \to 0} \frac{\vec{R}(x + \Delta x, y) - \vec{R}(x, y)}{\Delta x} \tag{2}$$

In Cartesian coordinates this derivative has length one and points along the x-axis, so it coincides with the usual basis vector:

$$\vec{e_x} \equiv \frac{\partial \vec{R}}{\partial x} \tag{3}$$

This reasoning can be easily applied to the other direction, but generalized further to any coordinate system.

Important: dividing by Δx does *not* normalize the vector in general. The length of $\partial \vec{R}/\partial c$ tells us how much distance we travel per unit change of the coordinate c, and this is only equal to 1 for special coordinates like the Cartesian ones. We'll see below that in polar coordinates $\partial \vec{R}/\partial \theta$ has length r.

The neat thing about this construction is that you can do it for any position vector, and you will still get your basis vectors. Fundamentally, this trick work because, in obtaining a slightly off position vector, a small change in coordinate is added in one coordinate only while keeping other coordinates constant. Then, we divide the resultant vector (obtained from the difference between the original position vector and the new slightly off one) by that small change, to get a rate of change that doesn't depend on the step size.

Now you should remember when we discussed, in the other notes for "Tensors for Beginners" about the forward and backward transformations when we move from one basis vector coordinate system to another.

Turns out that those transformation rules can be obtained easily once we introduced this above concept about basis vectors being partial derivatives of the position vector with respect to the coordinate variables.

And also, that these transformation matrices have a proper name: **the Jacobian matrices**

Let's say we want to find the forward transformation matrix (the Jacobian) matric when moving from cartesian to polar coordinates. This would be just a matter of applying the chain rule:

$$\vec{\tilde{e}_r} = \frac{\partial \vec{R}}{\partial r} = \frac{\partial \vec{R}}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial \vec{R}}{\partial y}\frac{\partial y}{\partial r} = \frac{\partial x}{\partial r}\vec{e_x} + \frac{\partial y}{\partial r}\vec{e_y}$$

$$\vec{\tilde{e}_\theta} = \frac{\partial \vec{R}}{\partial \theta} = \frac{\partial \vec{R}}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial \vec{R}}{\partial y}\frac{\partial y}{\partial \theta} = \frac{\partial x}{\partial \theta}\vec{e_x} + \frac{\partial y}{\partial \theta}\vec{e_y}$$

We know the transformation rules between cartesian and polar coordinates:

$$x = r\cos\theta, \qquad y = r\sin\theta.$$

so it's a matter of taking the derivatives to find that the Jacobian matrix, or forward matrix is:

$$\frac{\partial \vec{R}}{\partial r} = \frac{\partial \vec{R}}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial \vec{R}}{\partial y}\frac{\partial y}{\partial r} = \cos\theta \frac{\partial \vec{R}}{\partial x} + \sin\theta \frac{\partial \vec{R}}{\partial y}$$
$$\frac{\partial \vec{R}}{\partial \theta} = \frac{\partial \vec{R}}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial \vec{R}}{\partial y}\frac{\partial y}{\partial \theta} = -r\sin\theta \frac{\partial \vec{R}}{\partial x} + r\cos\theta \frac{\partial \vec{R}}{\partial y}$$

So:

$$F = \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{bmatrix} = \begin{bmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{bmatrix} \tag{4}$$

Notice that the second column has length r:

$$\tilde{\vec{e_\theta}} = \frac{\partial \vec{R}}{\partial \theta} = -r\sin\theta\, \vec{e_x} + r\cos\theta\, \vec{e_y} = r\,\hat{e}_\theta$$

So the coordinate basis vector $\tilde{\vec{e_\theta}}$ is **not** the unit vector $\hat{e}_\theta$ defined at the start of this section: it is r times longer, because a unit change in θ moves us a distance r along the circle. Only $\tilde{\vec{e_r}} = \hat{e}_r$ has unit length. This is exactly why the polar metric tensor has $\tilde{g}_{\theta\theta} = r^2$, as we'll see later.

And you can tackle, with the same derivation logic, the inverse Jacobian matrix, of backward transformation matrix, to obviously find out that the multiplication of the two yields the identity matrix.

Now, following a more tensorial approach, we can also re-write these transformation rules as:

$$\frac{\partial \vec{R}}{\partial p^i} = \frac{\partial \vec{R}}{\partial c^j}\frac{\partial c^j}{\partial p^i}$$
$$\frac{\partial \vec{R}}{\partial c^i} = \frac{\partial \vec{R}}{\partial p^j}\frac{\partial p^j}{\partial c^i}$$

3 Vectors as derivatives and their transformation rules

Allright we saw that basis vectors can be re-interpreted as partial derivative of the position vector along the coordinate axis.
We can sort of extend this logic to the vectors themselves. We saw that figuring out vector components in a particular basis is kind of easy to do for individual vectors, but now we're going do it for vector fields (i.e. a vector for every point in space), **but for now, we're going to deal with a slightly different type of vector field, and that's a vector field along a curve, so basically we're defining a vector at every point along a curve.**

The way we imagine vector fields along a curve is considering the set of tangent vectors to the curves. This gives us a vector field defined at every point on the curve.

A curve can be tought as a function that takes some input parameter λ and outputs the position vector $\mathbf{R}$ that traces our curve in the space.
If we want to have a vector tangent to this curve, we just need to consider taking the derivative of the position vector with respect to its parameter:

$$\frac{\mathrm{d}\vec{R}}{\mathrm{d}\lambda} \tag{5}$$

Now you may think that finding out the components of this vector in a specific basis can be difficult but it is really not, if we make use of the multivariable chain rule and expand it out in the coodinates we're working in.
For example, in the cartesian coordinates c^i and using the Einstein summation convention notation:

$$\frac{\mathrm{d}\vec{R}}{\mathrm{d}\lambda} = \frac{\mathrm{d}c^i}{\mathrm{d}\lambda}\frac{\partial\vec{R}}{\partial c^i} \tag{6}$$

At the end of the day, considering doing it in cartesian $\mathbb{R}^{\not\vdash}$ for simplicity, these are equivalent (individual vector on the top, and vector field along a curve on the bottom):

$$\vec{v} = v^1\vec{e_1} + v^2\vec{e_2}$$
$$\frac{\mathrm{d}\vec{R}}{\mathrm{d}\lambda} = \frac{\mathrm{d}x}{\mathrm{d}\lambda}\frac{\partial\vec{R}}{\partial x} + \frac{\mathrm{d}y}{\mathrm{d}\lambda}\frac{\partial\vec{R}}{\partial y}$$

The same can be applied to whatever coordinate system we're working in, so to give you an example, let's say we have the circular curve of radius 2, centered at the origin in $\mathbb{R}^{\not\vdash}$, defined by:

$$\vec{R}(\lambda) = (x(\lambda), y(\lambda)) = (2\cos\lambda, 2\sin\lambda) \tag{7}$$

The vector field along the curve is made up of the tangent vectors along the curve, that can be expanded as defined:

$$\begin{aligned}\frac{\mathrm{d}\vec{R}}{\mathrm{d}\lambda} &= \frac{\mathrm{d}x}{\mathrm{d}\lambda}\frac{\partial\vec{R}}{\partial x} + \frac{\mathrm{d}y}{\mathrm{d}\lambda}\frac{\partial\vec{R}}{\partial y} \\ &= -2\sin\lambda\frac{\partial\vec{R}}{\partial x} + 2\cos\lambda\frac{\partial\vec{R}}{\partial y}\end{aligned}$$

The components can also be written, for the cartesian basis, as a vector column:

$$\begin{bmatrix}-2\sin\lambda \\ 2\cos\lambda\end{bmatrix}_{\frac{\partial}{\partial c^i}} \tag{8}$$

We can do the same in polar coordinates r, θ, where the same curve is defined as:

$$\vec{R}(\lambda) = (r(\lambda), \theta(\lambda)) = (2, \lambda) \tag{9}$$

$$\begin{aligned}\frac{\mathrm{d}\vec{R}}{\mathrm{d}\lambda} &= \frac{\mathrm{d}r}{\mathrm{d}\lambda}\frac{\partial\vec{R}}{\partial r} + \frac{\mathrm{d}\theta}{\mathrm{d}\lambda}\frac{\partial\vec{R}}{\partial\theta} \\ &= 0\frac{\partial\vec{R}}{\partial r} + 1\frac{\partial\vec{R}}{\partial\theta}\end{aligned}$$

So:

$$\begin{bmatrix}0\\1\end{bmatrix}_{\frac{\partial}{\partial p^i}} \tag{10}$$

Now, as you can already imagine, like we saw that for individual vectors, the components are contravariant (i.e. they transform the opposite way of the basis vectors), we're going to simply show that this is valid indeed in this vector fields case.

We already saw in the previous section, the transformation rules of the basis vectors using the Jacobain and Inverse Jacobian:

$$\frac{\partial\vec{R}}{\partial p^i} = \frac{\partial\vec{R}}{\partial c^j}\frac{\partial c^j}{\partial p^i} \qquad \text{Forward Transform} \tag{11a}$$

$$\frac{\partial\vec{R}}{\partial c^i} = \frac{\partial\vec{R}}{\partial p^j}\frac{\partial p^j}{\partial c^i} \qquad \text{Backward Transform} \tag{11b}$$

Now we can just use these and replace the basis vectors with their new basis transformation rule:

$$\frac{\mathrm{d}\vec{R}}{\mathrm{d}\lambda} = \frac{\mathrm{d}c^i}{\mathrm{d}\lambda}\frac{\partial\vec{R}}{\partial c^i} = \frac{\mathrm{d}c^i}{\mathrm{d}\lambda}\left(\frac{\partial\vec{R}}{\partial p^j}\frac{\partial p^j}{\partial c^i}\right) = \left(\frac{\mathrm{d}c^i}{\mathrm{d}\lambda}\frac{\partial p^j}{\partial c^i}\right)\frac{\partial\vec{R}}{\partial p^j} \tag{12}$$

This is a linear combination of the new basis vectors $\frac{\partial\vec{R}}{\partial p^j}$, and since we know that we can write:

$$\frac{\mathrm{d}\vec{R}}{\mathrm{d}\lambda} = \frac{\mathrm{d}p^i}{\mathrm{d}\lambda}\frac{\partial\vec{R}}{\partial p^i} \tag{13}$$

This means that:

$$\frac{\mathrm{d}p^i}{\mathrm{d}\lambda} = \frac{\mathrm{d}c^j}{\mathrm{d}\lambda}\frac{\partial p^i}{\partial c^j} \tag{14}$$

Which shows that the vector field components transform using the backward transformation (inverse Jacobian)

We can actually test this out in the previous example right? So we have the components being, in both coordinate systems:

$$\begin{bmatrix} -2\sin\lambda \\ 2\cos\lambda \end{bmatrix}_{\frac{\partial}{\partial c^i}} \qquad\qquad \begin{bmatrix} 0 \\ 1 \end{bmatrix}_{\frac{\partial}{\partial p^i}}$$

And we also know the Inverse Jacobian in the case of polar-to-cartesian coordinates, being:

$$B = J^{-1} = \begin{bmatrix} x/r & y/r \\ -y/r^2 & x/r^2 \end{bmatrix} = \begin{bmatrix} \cos\lambda & \sin\lambda \\ -1/2\sin\lambda & 1/2\cos\lambda \end{bmatrix} \tag{15}$$

Let's check that:

$$\begin{aligned} B\begin{bmatrix} -2\sin\lambda \\ 2\cos\lambda \end{bmatrix}_{\frac{\partial}{\partial c^i}} &= \begin{bmatrix} \cos\lambda & \sin\lambda \\ -1/2\sin\lambda & 1/2\cos\lambda \end{bmatrix}\begin{bmatrix} -2\sin\lambda \\ 2\cos\lambda \end{bmatrix}_{\frac{\partial}{\partial c^i}} \\ &= \begin{bmatrix} -\cos\lambda 2\sin\lambda + \sin\lambda 2\cos\lambda \\ -1/2\sin\lambda(-2\sin\lambda) + 1/2\cos\lambda 2\cos\lambda \end{bmatrix}_{\frac{\partial}{\partial p^i}} \\ &= \begin{bmatrix} 0 \\ 1 \end{bmatrix}_{\frac{\partial}{\partial p^i}} \end{aligned}$$

Same can be checked on the opposite direction applying the forward/jacobian transformation.

One last thing to mention before terminating this section is in terms of notation: you notice that the position vector appears in a lot of them, so what we're going to do from here onward, is get rid of the position vector symbol.
This might seem like a weird choice of notation, but from now on, for example, the $\mathbb{R}^2$ cartesian basis vectors will be written as:

$$\vec{e_x} = \frac{\partial}{\partial x}$$
$$\vec{e_y} = \frac{\partial}{\partial y}$$

where basically we consider basis vectors these partial derivative operators.
If you think about this, it's not very weird at the end, as derivative operators are linear, as they can be scaled and added like vectors do, so they have both a magnitude and a direction. They also obey the forward and backward transformation rules that we would expect vectors to obey.

The main logical reason to get rid of the position vector symbol is that, the position vector definition requires an origin to be set, and later on when we're going to deal with

curved surfaces and manifolds, it's not easy to define an origin point.
So, using a position vector without an origin does not make a lot of sense, but we're still going to be able to use vectors and vector fields as long as we have these derivative operators to work with.

The whole main idea is the following:
Basis transforms with the Jacobian;
Components transform with the inverse Jacobian;
Their product gives the identity;
Therefore the geometric vector is coordinate-invariant.

$$\frac{\mathrm{d}}{\mathrm{d}\lambda} = \frac{\mathrm{d}p^i}{\mathrm{d}\lambda}\frac{\partial}{\partial p^i} = \frac{\mathrm{d}c^i}{\mathrm{d}\lambda}\frac{\partial}{\partial c^i} \qquad \text{Total derivative along a curve (chain rule)} \tag{16a}$$

$$\frac{\partial}{\partial p^i} = \frac{\partial}{\partial c^j}\frac{\partial c^j}{\partial p^i} \qquad \text{Coordinate basis transformation} \tag{16b}$$

$$\frac{\partial}{\partial c^i} = \frac{\partial}{\partial p^j}\frac{\partial p^j}{\partial c^i} \qquad \text{Inverse coordinate basis transformation} \tag{16c}$$

$$\frac{\mathrm{d}p^i}{\mathrm{d}\lambda} = \frac{\mathrm{d}c^j}{\mathrm{d}\lambda}\frac{\partial p^i}{\partial c^j} \qquad \text{Contravariant component transformation} \tag{16d}$$

$$\frac{\mathrm{d}c^i}{\mathrm{d}\lambda} = \frac{\mathrm{d}p^j}{\mathrm{d}\lambda}\frac{\partial c^i}{\partial p^j} \qquad \text{Inverse component transformation} \tag{16e}$$

3.1 Hold on - Why are derivative operators vectors?

Let's here make some clarity about what we've discussed so far in the previous two sections. We brought up this new way of thinking about basis vectors, where we went from considering basis vectors as derivative of the position vector wrt the coordinate variables, to them being defined by the partial derivative operators themselves.
Let's start by saying that this way of thinking will be appreciated more when we'll start thinking anout curved (differential) geometry, but we chose to introduce it right now, even though we're still working in flat space, to already start using these ideas and notations beforehand.

We just a bit ahead of ourselves here, to explain the main reason of already working with this, so you get an idea of what we're going to deal with, and then we'll get back to our flat space for now.

We saw that if we consider the difference between two position vectors, a small distance h apart from each other in the direction of one coordinate axis, divide this by this distance h, and consider the limit when h approaches 0, gives us as result the basis vector along the considered coordinate direction (which has unit length only in special coordinates like Cartesian ones). We said that this position vector derivative along the coordinate is what we take as the basis vector.

We also shown that, for tangent vectors along a curve, parametrized by the paramter λ, considering the position vector at a certain position λ and another one a small step h away, we can compute the tangent vector to this curve at position λ, by taking the difference and consider the limit when h approaches 0.

And we can expand the tangent vector out as a linear combination of the basis vectors using the chain rule. Now all of this, has been discussed working on the flap 2D plane.

Let's try to move further and consider this in a curved space, for example the surface of the earth, to see how the situation is different.

If we consider a point moving on the surface of the earth from one position to another, and we want to compute its velocity, we need to find the tangent vector to the curve he moves along. So what do we do? We can take the derivative of the position vector wrt to the parameter we use to define this curve.
Okay, so how do we define the position vector? We need to define an origin, right? We can think of putting this origin at the center of the sphere (earth)....but this is a bit of a different situation than before when working in the flat space?

Why? Well first because the origin does not live on the earth's 2D surface, and second because the velocity vector itself, also does not live on the earth's 2D surface, since it extends to the outside space.
They both are external to the surface we're studying, and indeed this is referred to "extrinsic geometry"

Now, this may not seem a problem, if we consider the 3D surface of a globe, from an external perspective of a 3D space.
But what if instead of this situation, we're forced to look at the surface of the earth on a map?

How do we define position vectors here? There's not really a convenient outside place where we can put the origin here, there's no external place for us to use. You might think that you can potentially use any place on the map to put the origin, choosing a random placefor it and drawing a straight line representing the position vector.

There's a problem with this approach though, vectors are supposed to be straight, but the path from the origin to the tip of the vector is not really straight, because the surface in reality is not flat.

The method of studying curved surfaces where we're not allowed to leave the surface is called intrinsic geometry.

When we'll study General relativity, we're going to deal with a 4-dimensional curved spacetime geometry, the situation we're running on is exactly this. Spacetime is curved so we cannot really run straight lines, hence we cannot draw position vectors.

And we cannot pick an origin point outside of this 4-dim space, because that would mean picking a point outside universe, and that's completely meaningless.

So the summary is that, we're forced to use intrinsic geometry in General Relativity, so how are we supposed to define things like velocity, acceleration etc if we're not allowed to use position vectors?

The solution is, to realize that, while we cannot draw straight lines on a curved surface, we can still draw a curved path, and on a map, we can use grid lines, so we can still define derivatives wrt these curves. So we can use these derivative operators to talk about different directions on the surface.
The partial derivative wrt to x for example, points to the x direction, the partial wrt to y points to the y direction and so on.

If we have some path traveling around in our curved space, we can still consider the direction that the path is pointing to, using the derivative wrt the curve parameter used, and we break this direction up in terms of our x, y components, using the multi-variable chain rule as we saw.

Formally, our new notation with basis vectors defined as partial derivative operators along the axis directions, form a vector space called the "Tangent Vector Space" T_pM, which is the vector space of derivatives at some point p on a surface M

4 Differentials as covectors and their transformation rules

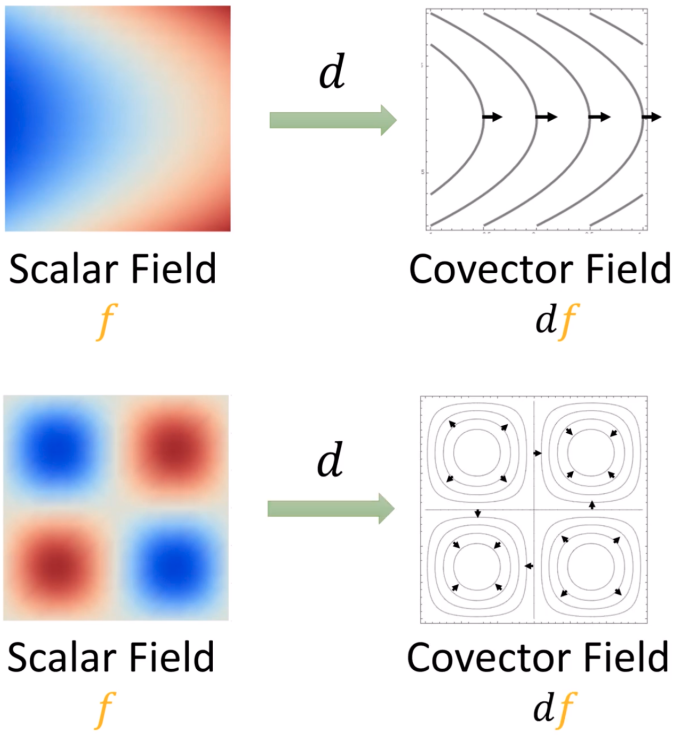
Let's start this chapter by reminind ourselves what differentials are and covectors are. We know that differentials in Integrals, like dx are small changes in the variable of integration, in this case it's a small change in the variable x.

Covectors instead, are these functionals acting on vectors, and outputting scalars, which can be visualized as stacks of lines that indeed eat a vector and produce a scalar. The scalar can be tought as the number of times the vector pierces the stack of lines.

So like, how can these two concepts be linked together, and actually how the hell are they the same thing?

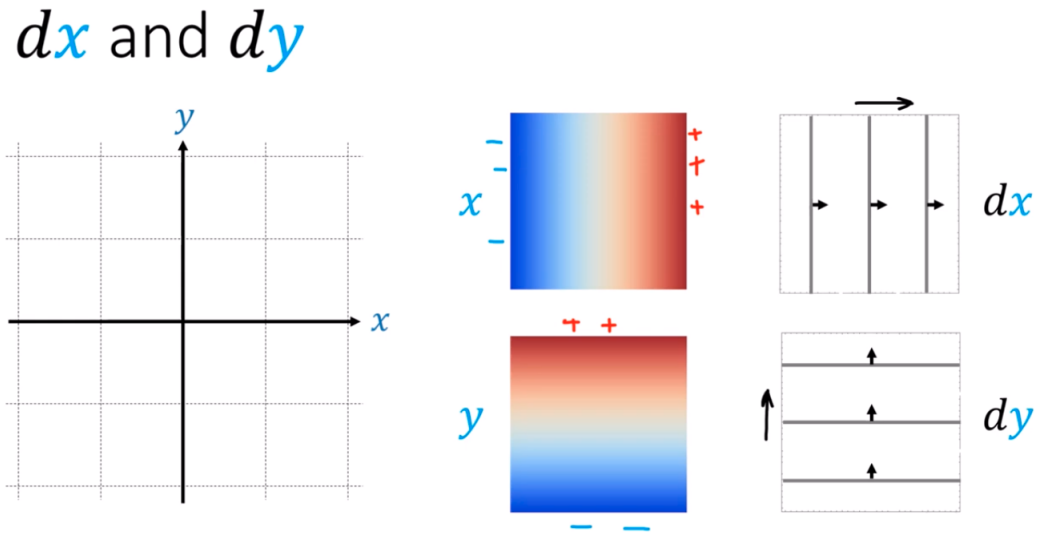
The key of understanding this, is the re-interpretation of the d symbol of the differentials. The old way of thinking about it, is that this symbol takes the variable and gives a small change of this variable.
The new interpretation is that d is an operator that takes a scalar field f and outputs a df covector field. A scalar field is a field where a scalar f is defined at every point in space, a covector field df is a field where a covector is defined at every point in space.

So how does this d operator work? A couple of pictures are worth a thousand words here:

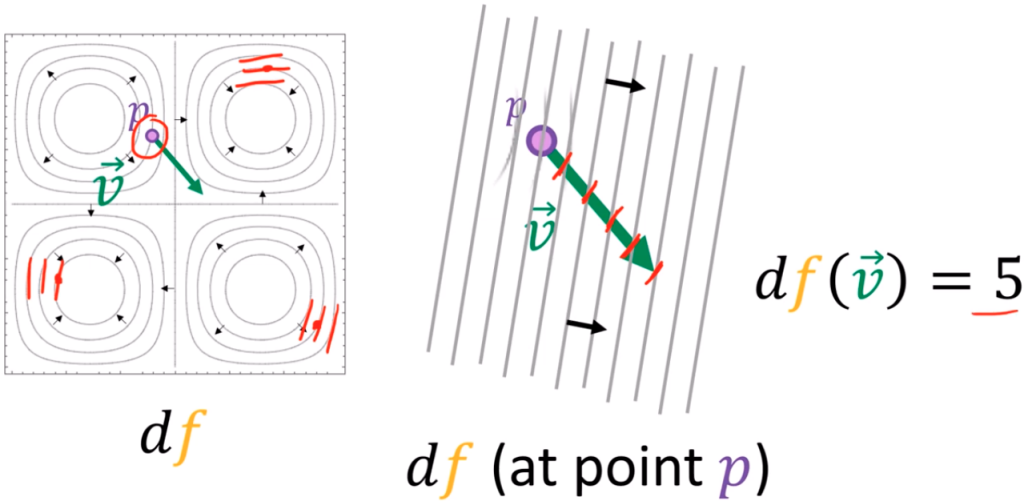


So like, the operator d takes a function and outputs the function's level sets. Sometimes, they're called 0-forms (functions) and 1-forms (level sets).

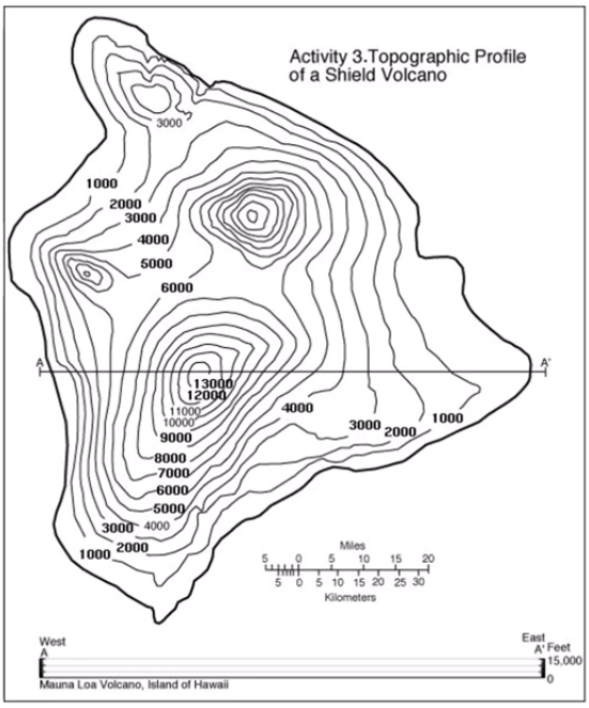
Let's give another example, showing the covector fields of the cartesian coordinates dx and dy:



Now, if this operator is really a covector field, we know that covectors eat vectors to output scalars, right? How does this work considering what we've seen so far?



And it's also quite straight forward to show that covector fields are linear. But whet's the geometrical meaning of $df(\vec{v})$



We can think of these level sets of a map as a covector field.
When acting on a vector, $df(\vec{v})$ is proportional to the steepness of f in the direction of $\vec{v}$ and also proportional to the length of $\vec{v}$.

So: $df(\vec{v})$ **tells us the rate of change of** f **when moving at velocity** $\vec{v}$

But then, this reminds us of the definition of directional derivative, isn't it? At the end of the day, $df(\vec{v})$ is the **directional derivative** of the function in the vector direction.

$$df(\vec{v}) = D_{\vec{v}}f = \frac{\partial f}{\partial \vec{v}} \tag{17}$$

4.1 Differential forms components and their transformation

Just as we can write vectors as linear combinations of special basis vectors, scaled by some components, we can break covector fields, or differential forms, as linear combination of some special differential forms basis scaled by their components.

$$df = Adx + Bdy \tag{18}$$

Let's start by checking how differential forms act on our special basis vectors $\frac{\partial}{\partial x}, \frac{\partial}{\partial y}$:

$$df(\vec{e_x}) = df\left(\frac{\partial}{\partial x}\right) = \frac{\partial f}{\partial x} \tag{19}$$

That's the directional derivative of f in the x direction.
Same for the other direction:

$$df(\vec{e_y}) = df\left(\frac{\partial}{\partial y}\right) = \frac{\partial f}{\partial y} \tag{20}$$

Now we can look at the simple scalar field x, and the scalar value at a point is just the value of x at that point, positives on the right, negative on the left.
The corresponding covector field is dx, represented by vertical level sets directed to the right.
So:

$$\begin{aligned} dx(\vec{e_x}) &= dx\left(\frac{\partial}{\partial x}\right) \\ &= \frac{\partial x}{\partial x} = 1 \\ dx(\vec{e_y}) &= dx\left(\frac{\partial}{\partial y}\right) \\ &= \frac{\partial x}{\partial y} = 0 \end{aligned}$$

Same process can be applied to the scalar field y, obtaining in summary:

$$dx\left(\frac{\partial}{\partial x}\right) = \frac{\partial x}{\partial x} = 1 \qquad dx\left(\frac{\partial}{\partial y}\right) = \frac{\partial x}{\partial y} = 0$$
$$dy\left(\frac{\partial}{\partial x}\right) = \frac{\partial y}{\partial x} = 0 \qquad dy\left(\frac{\partial}{\partial y}\right) = \frac{\partial y}{\partial y} = 1$$

Which can be re-written, considering $c^1 = x, c^2 = y$:

$$dc^i\left(\frac{\partial}{\partial c^j}\right) = \frac{\partial c^i}{\partial c^j} = \delta^i_j \tag{21}$$

As when we discussed the special dual basis (co)vectors in the "Tensor for Beginners" notes, these special differential forms dx, dy are the dual basis for differential forms.
To find the coefficients, let's just consider a generic function $f(x, y)$ and a curve parametrized by the parameter λ.
The tangent vectors along the curve are given by $\frac{\partial}{\partial\lambda}$.
Let's consider the rate of change of this function as we travel along the curve, which is given by the derivative $\frac{df}{d\lambda}$

Now remember this, the covector field df, acting on the vector $d/d\lambda$ is equal to $df/d\lambda$ so we can re-write:

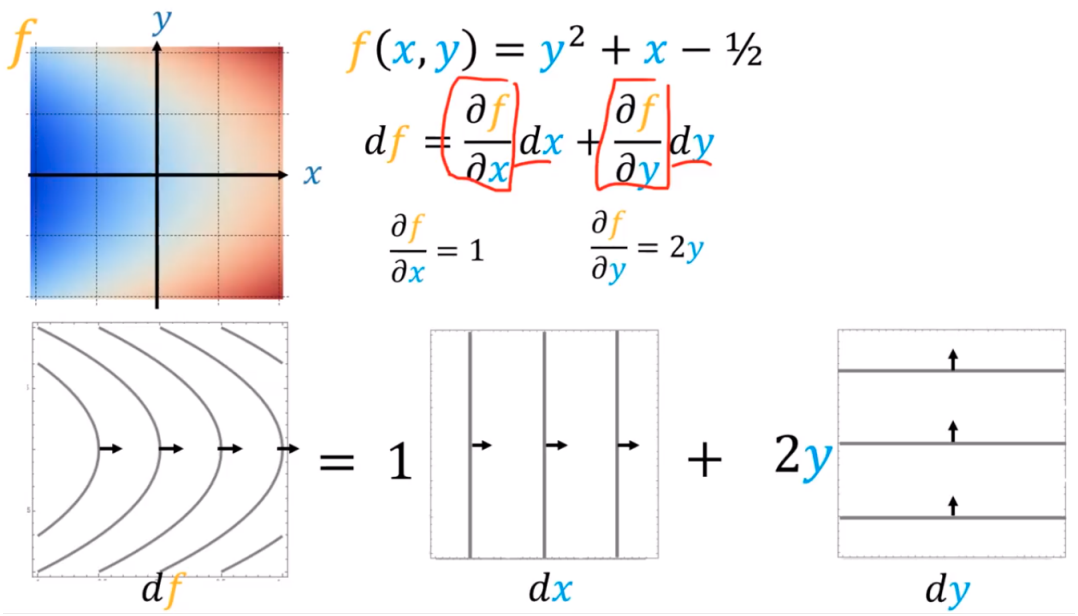
$$\begin{aligned}
\frac{df}{d\lambda} &= \frac{\partial f}{\partial x}\frac{dx}{d\lambda} + \frac{\partial f}{\partial y}\frac{dy}{d\lambda} \\
df\left(\frac{d}{d\lambda}\right) &= \frac{\partial f}{\partial x}dx\left(\frac{d}{d\lambda}\right) + \frac{\partial f}{\partial y}dy\left(\frac{d}{d\lambda}\right) \\
df\left(\frac{d}{d\lambda}\right) &= \left(\frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy\right)\left(\frac{d}{d\lambda}\right)
\end{aligned}$$

So from the last equation, we can deduce that these two quantities are the same vector field:

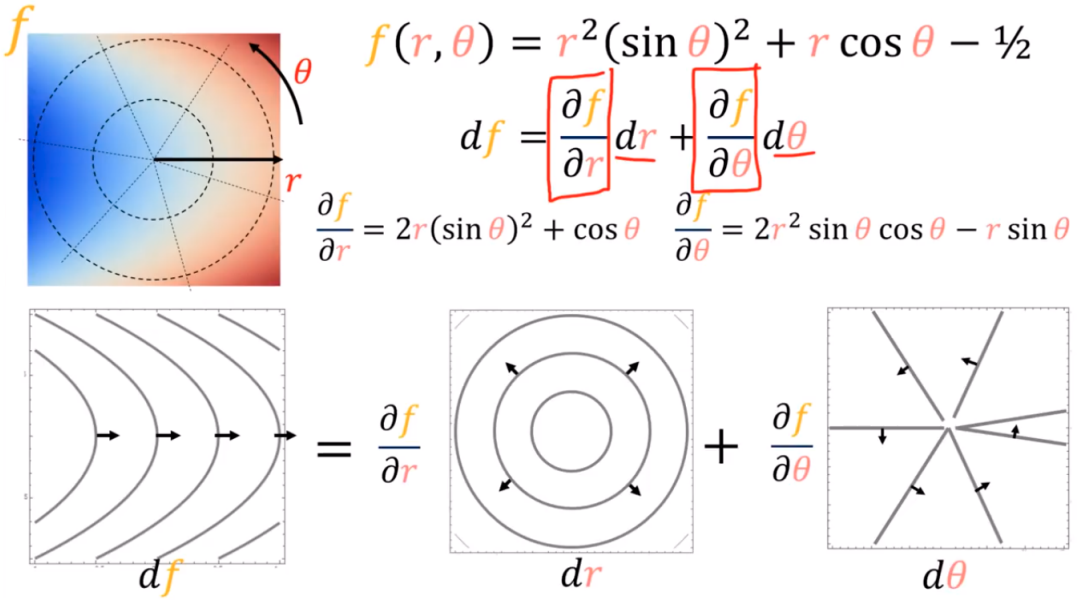
$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy \tag{22}$$

And this is indeed what we were looking for, writing the covector field as linear combination of dx, dy, so dx, dy form a basis for the set of covector fields. They're the dual basis for differential forms.

Let's see a more concrete example to understand this better. Consider this scalar field, we can write down the covector field in terms of the basis vectors in this way:



The same can be done in other coordinate systems, like polar ones:



Now, how would both the covector basis field and the covector field components transform? Well we just showed that any covector field can be written as linear combination of basis covector field dx and dy.

But since this is valid for any covector field, we can expand our new polar covector field basis:

$$dr = \frac{\partial r}{\partial x}dx + \frac{\partial r}{\partial y}dy$$
$$d\theta = \frac{\partial \theta}{\partial x}dx + \frac{\partial \theta}{\partial y}dy$$

If you recall, those coefficients are just the entries of the Inverse Jacobian Matrix B or J^{-1}. What this means is that, covector basis field transform contravariantly.
In Einstein notation, and then in matrix notation, we can write:

$$dp^i = \frac{\partial p^i}{\partial c^j}dc^j$$
$$\begin{bmatrix} dr \\ d\theta \end{bmatrix} = \begin{bmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \end{bmatrix} \begin{bmatrix} dx \\ dy \end{bmatrix}$$

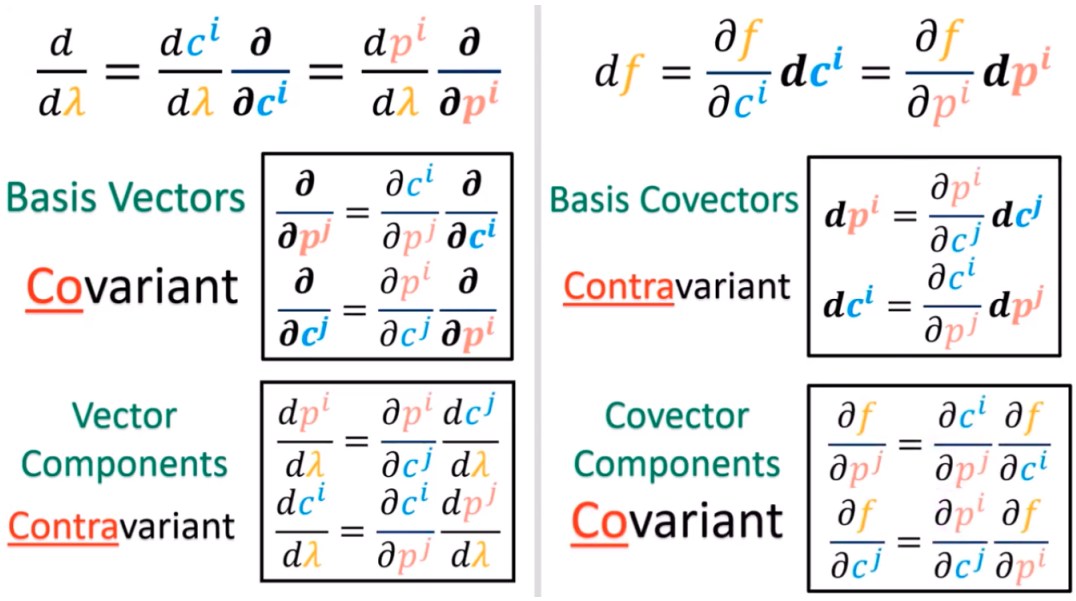
Remember that when writing in matrix notation, to make formulas work, we need to write contravariant components as column vectors.
The other way around is also possible, so we'd have that to go from the new to the hold covector basis field, the Jacobian, or forward matrix is used:

$$dc^i = \frac{\partial c^i}{\partial p^j}dp^j$$
$$\begin{bmatrix} dx \\ dy \end{bmatrix} = \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{bmatrix} \begin{bmatrix} dr \\ d\theta \end{bmatrix}$$

Now what about the covector field components?
Quite straight forward just using the usual chain rule:

$$\frac{\partial f}{\partial p^j} = \frac{\partial c^i}{\partial p^j}\frac{\partial f}{\partial c^i}$$
$$\frac{\partial f}{\partial c^j} = \frac{\partial p^i}{\partial c^j}\frac{\partial f}{\partial p^i}$$

So while the basis covector fields obey the contravariant transformation rule, the covector field components obey the covariant one.
A full summary is depicted below:



5 Integration with differential forms

What we've discussed so far has shown that out operator d takes a scalar field f and produces a covector field df which is the level sets of f oriented towards the positive values of f.
And this covector field df can behave like a function that acts on vectors.
If applied to a vector $\vec{v}$, we think about zooming on the covector at the point of $\vec{v}$ and count how many times the vector piereces to get the output.

So this reinterpretation of the operator d gives us covector fields which are also called differential forms. How does this relate back to the differential we see in integrals, which represents a small change in the varibale, for example dx?

The short answer is that, every single integral involves:

- a differential form (covector field)
- a path

$$\int_a^b f(x)dx \tag{23}$$

The result of the integral is just the number of covector stacks pierced by the path.

To prove this, we're going to explore a couple of examples from Physics, but first let's quickly review two important theorems from calculus:

- Fundamental Theorem of Calculus

$$\int_a^b f(x)dx = F(b) - F(a)$$
$$\int_a^b \frac{dF}{dx}dx = F(b) - F(a)$$

- Gradient Theorem

$$\int_{P[a,b]} \nabla F \cdot dr = F(b) - F(a)$$

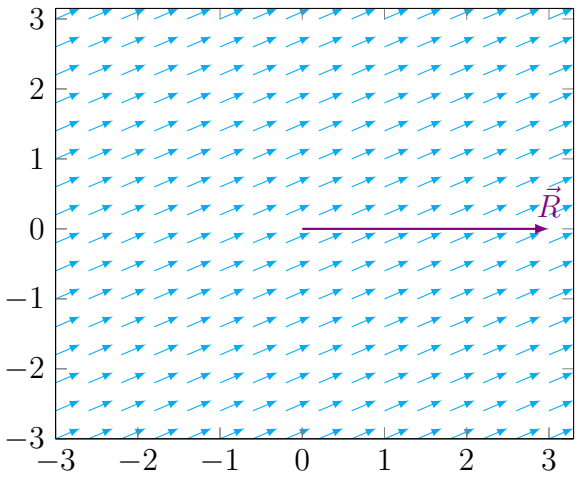
Both of them involve integrals over a path from a to b, where the integral itself only depends on the values at the endpoints.
We're going to prove the Gradient theorem to show why differential forms are strictly tied to integrals.

Recall what work is in physics, it's the application of a force to an object along some distance, the force dot product with the displacement vector:

$$W = \vec{F} \cdot \vec{R} \tag{24}$$

The dot product actually tells us how much of the force was used to move the object along the displacement vector.
Let's say that we have some force vector defined as:

$$\vec{F} = 2\vec{e_1} + 1\vec{e_2} \tag{25}$$



And our vector giving us the direction is:

$$\vec{R} = 3\vec{e_1} + 0\vec{e_2} \tag{26}$$

The total work would be $W = \vec{F} \cdot \vec{R} = 6$ Joules

Now, for a bit more meaningful example, consider the force field due to the gravitational field that the earch exercit on a smaller mass m, at a distance $\vec{R}$:

$$\vec{F} = \frac{GMm}{\|\vec{R}\|^2}(-\vec{e_r}) \tag{27}$$

Giving a curved path with an object traveling along, how do we compute the work done on this object, by the gravitational field. The formula we used before is not valid here, as the force field is different at every point in space, and the path we're traveling along is curved.

We actually need to perform an integration along the path like:

$$W = \int_P \vec{F}_G \cdot d\vec{R} \tag{28}$$

This $d\vec{R}$, what it really is, it's just a short way of seeing $\frac{d\vec{R}}{d\lambda}d\lambda$.
Where λ is like the time parameter that increases as we move along the curve. So really, $\frac{d\vec{R}}{d\lambda}$ is the velocity vector or the tangent vector along the path.

$$W = \int_P \vec{F}_G \cdot \frac{d\vec{R}}{d\lambda}d\lambda \tag{29}$$

Let's compute this for a couple of different paths, circular and straight out from earth:

Circular path $\vec{R}(\lambda) = (r = 2, \theta = \lambda)$

$$\frac{d\vec{R}}{d\lambda} = \frac{dr}{d\lambda}\frac{d\vec{R}}{dr} + \frac{d\theta}{d\lambda}\frac{d\vec{R}}{d\theta}$$
$$\frac{d\vec{R}}{d\lambda} = 0\vec{e_r} + 1\vec{e_\theta}$$
$$W = \int_P \left(\frac{GMm}{\|\vec{R}\|^2}(-\vec{e_r})\right) \cdot (\vec{e_\theta})d\lambda = 0$$

Because the two basis vectors are orthogonal to each other.
This means that over a circular path with constant radius, no work gets done by the gravitational field, as there's no component of its force along the curve.

Straight path ourwards from earth $\vec{R}(\lambda) = (r = \lambda, \theta = 0)$

$$\frac{d\vec{R}}{d\lambda} = 1\vec{e_r} + 0\vec{e_\theta}$$
$$W = \int_P \left(\frac{GMm}{\|\vec{R}\|^2}(-\vec{e_r}) \right) \cdot (\vec{e_r})d\lambda$$
$$= \int_a^b -\frac{GMm}{\lambda^2}d\lambda = GMm \left[\frac{1}{\lambda}\right]_a^b$$
$$= GMm \left(\tfrac{1}{b} - \tfrac{1}{a}\right)$$

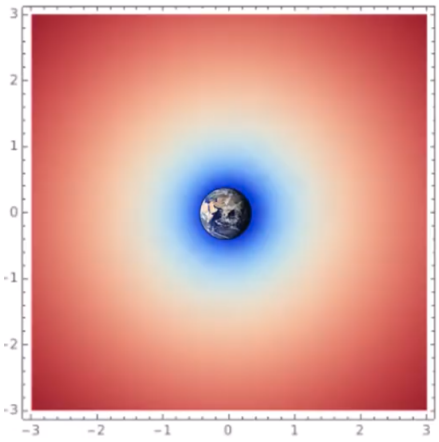
If we stop a moment to think, and look at the geometric picture of it, it's kind of not very intuitive that, whatever path we take, the integral always depend only on the endpoints (i.e. start and finish) of the path, even though we're actually considering all those tangent vectors along the path, and computing the dot product with the gravitational field.
It's pretty amazing, if not magical, that all those add up so that the total amount is only dependent on the endpoints, whatever strange curve we go along between points a and b, isnt' it?

There's indeed a better way to visualize this, that also makes this fact, that the work only depends on the start and end of the path, very intuitive and also kind of obvious.

To talk about this, we need to introduce the gravitational potential.
First we introduce the gravitational field, which is nothing more than the gravitational force we saw before, but that only depends on the source mass generating it, so without the small mass m.
This vector field can be written as the gradient of a scalar field, that we call gravitational potential:

$$\vec{F_G} = m\vec{G}$$
$$\vec{G} = -\boldsymbol{\nabla}\Phi$$
$$\vec{F_G} = m(-\boldsymbol{\nabla}\Phi)$$

The gravitational potential can be visualized as:



Now using this definition of the gravitational force as dependning on the gradient of the gravitational potential, we can write, considering cartesian coordinates:

$$
\begin{aligned}
W &= \int_P \vec{F_G} \cdot \frac{d\vec{R}}{d\lambda} d\lambda \\
&= \int_P m(-\boldsymbol{\nabla}\Phi) \cdot \frac{d\vec{R}}{d\lambda} d\lambda \\
\boldsymbol{\nabla}\Phi &= \frac{\partial \Phi}{\partial x}\vec{e_x} + \frac{\partial \Phi}{\partial y}\vec{e_y} \\
\frac{d\vec{R}}{d\lambda} &= \frac{dx}{d\lambda}\frac{d\vec{R}}{dx} + \frac{dy}{d\lambda}\frac{d\vec{R}}{dy} \\
&= \frac{dx}{d\lambda}\vec{e_x} + \frac{dy}{d\lambda}\vec{e_y}
\end{aligned}
$$

$$
\begin{aligned}
&\boldsymbol{\nabla}\Phi \cdot \frac{d\vec{R}}{d\lambda} \\
&\quad = \frac{\partial \Phi}{\partial x}\frac{dx}{d\lambda} + \frac{\partial \Phi}{\partial y}\frac{dy}{d\lambda}
\end{aligned}
$$

So that:

$$
\begin{aligned}
W &= \int_P m(-\boldsymbol{\nabla}\Phi) \cdot \frac{d\vec{R}}{d\lambda} d\lambda \\
&= -m \int_P \frac{d\Phi}{d\lambda} d\lambda \\
&= -m \int_P d\Phi \\
&= -m \left[\Phi(P_{end}) - \Phi(P_{start})\right]
\end{aligned}
$$

So really, when we write down the integral with the dot product between the gradient and the tangent vectors along the curve, we're sort of overcomplicating things, because all

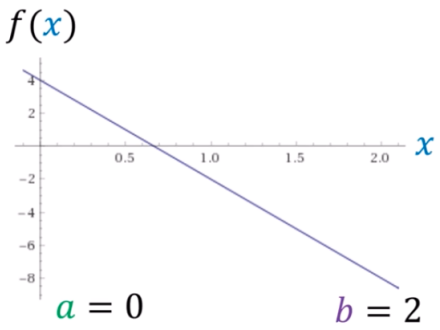
of it is just the expaneded version of $d\Phi$.
We just have to evaluate Φ at the end and at the start of the curve and then subtract.

Looking at the scalar field and not the associated vector field (given by the gradient of the scalar field) does indeed make more sense to compute the work.
Then, yet another way of looking at things, instead of looking at the scalar field Φ, we can look at the level sets of Φ, and when we draw a path through these level sets, if we can to compute the work along, we just look at where the path starts, where it ends, and see how much the function has changed. Basically by counting how many level sets, the path piereces from start to end.

We can move ahead with another of example, with a simple single variable integration.

$$\int_0^2 (-6x + 4)dx$$

Integration is over the single variable x. If we represent this function and compute the integral as we are usually doing:



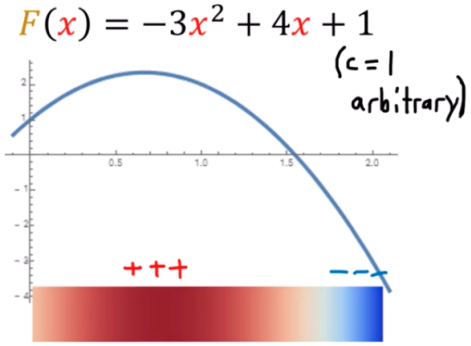
$$\begin{aligned}
&\int_0^2 (-6x + 4)dx \\
&= \left[-3x^2 + 4x\right]_0^2 \\
&= \left[-3(2)^2 + 4(2)\right] - \left[-3(0)^2 + 4(0)\right] \\
&= -12 + 8 \\
&= -4
\end{aligned}$$

We can think of re-interpreting this using our new covector field vision:

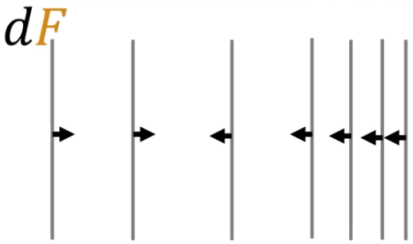
$$\int_0^2 (-6x+4)dx$$
$$= \int_0^2 \frac{dF}{dx}dx$$
$$= \int_0^2 dF$$

$$\frac{dF}{dx} = -6x + 4$$
$$F(x) = -3x^2 + 4x + C$$

Visualising it, it would look like this function here, but we can also visualize it as scalar field:



The associated covector field dF would look like this:



If you now plot the path from $x = 0$ to $x = 2$, you can clearly see yourself that, adding a $+1$ when the path pierces the covector field along its direction, and a -1 when on the opposite direction, the path crosses the covector field exactly $+1 - 1 - 1 - 1 - 1 - 1 = -4$ times, which is exactly the same value we got before using standard calculus.

6 The metric tensor and arc length - flat and curved spaces

We already saw in the *Tensor for beginners* notes, that the metric tensor is used to compute the length of a vector, the dot product between two vectors and the angle in between, and we also saw that it transforms using 2 covariant rules, hence is a (0,2)-tensor (bilinear form).

$$g : V \times V \to \mathbb{R}$$
$$g\left(\vec{v}, \vec{w}\right) \mapsto v^i w^j g_{ij}$$

$$\tilde{g_{ij}} = F_i^k F_j^l g_{kl}$$
$$g_{kl} = B_k^i B_l^j \tilde{g_{ij}}$$

We'll here focus on using the metric tensor to compute lengths of curves in flat space.

6.1 Arc length in flat spaces

We recall that in order to get the length of a curve in multi-variable calculus, basically we approximate the curve by a series of straight line segments, sum them up and take the limit as the Δ parameter used to parametrize the curve approaches to 0.
At the end of the day, the key to get the curve length is getting the magnitude of the vector $\frac{dR}{d\lambda}$ if λ is the parameter, and R the position vector used to describe a point on the curve. Which is nothing but the tangent velocity vector along the curve:

$$\text{arc length} = \int \left\| \frac{d\vec{R}}{d\lambda} \right\| d\lambda$$

Now, to get this magnitude, we can just compute the dot product of this vector with itself, as we know, and do that in cartesian coordinates for example:

$$\begin{aligned}
\left\| \frac{d\vec{R}}{d\lambda} \right\|^2 &= \frac{d\vec{R}}{d\lambda} \cdot \frac{d\vec{R}}{d\lambda} \\
&= \left(\frac{\partial \vec{R}}{\partial x}\frac{dx}{d\lambda} + \frac{\partial \vec{R}}{\partial y}\frac{dy}{d\lambda} \right) \cdot \left(\frac{\partial \vec{R}}{\partial x}\frac{dx}{d\lambda} + \frac{\partial \vec{R}}{\partial y}\frac{dy}{d\lambda} \right) \\
&= \left(\frac{dx}{d\lambda} \right)^2 \left(\frac{\partial \vec{R}}{\partial x} \cdot \frac{\partial \vec{R}}{\partial x} \right) + 2\frac{dx}{d\lambda}\frac{dy}{d\lambda} \left(\frac{\partial \vec{R}}{\partial x} \cdot \frac{\partial \vec{R}}{\partial y} \right) + \left(\frac{dy}{d\lambda} \right)^2 \left(\frac{\partial \vec{R}}{\partial y} \cdot \frac{\partial \vec{R}}{\partial y} \right)
\end{aligned}$$

This ugly long product chain will look similar in other coordinate systems, like polar:

$$= \left(\frac{dr}{d\lambda} \right)^2 \left(\frac{\partial \vec{R}}{\partial r} \cdot \frac{\partial \vec{R}}{\partial r} \right) + 2\frac{dr}{d\lambda}\frac{d\theta}{d\lambda} \left(\frac{\partial \vec{R}}{\partial r} \cdot \frac{\partial \vec{R}}{\partial \theta} \right) + \left(\frac{d\theta}{d\lambda} \right)^2 \left(\frac{\partial \vec{R}}{\partial \theta} \cdot \frac{\partial \vec{R}}{\partial \theta} \right)$$

If you carefully look at this, you will find out that it's very similar to when we computed the magnitude of a vector in our *Tensor for beginners* notes:

$$
\begin{aligned}
\|\vec{v}\|^2 &= \vec{v} \cdot \vec{v} \\
&= \left(v^1 \vec{e_1} + v^2 \vec{e_2}\right) \cdot \left(v^1 \vec{e_1} + v^2 \vec{e_2}\right) \\
&= \left(v^1\right)^2 \left(\vec{e_1} \cdot \vec{e_1}\right) + 2 v^1 v^2 \left(\vec{e_1} \cdot \vec{e_2}\right) + \left(v^2\right)^2 \left(\vec{e_2} \cdot \vec{e_2}\right) \\
&= \left(\tilde{v^1}\right)^2 \left(\tilde{\vec{e_1}} \cdot \tilde{\vec{e_1}}\right) + 2 \tilde{v^1} \tilde{v^2} \left(\tilde{\vec{e_1}} \cdot \tilde{\vec{e_2}}\right) + \left(\tilde{v^2}\right)^2 \left(\tilde{\vec{e_2}} \cdot \tilde{\vec{e_2}}\right)
\end{aligned}
$$

The key is indeed getting those dot products like we did in those initial notes, and those are nothing more than the components of the metric tensor in different coordinate systems.

$$
g_{c^i} = \begin{bmatrix} \frac{\partial \vec{R}}{\partial x} \cdot \frac{\partial \vec{R}}{\partial x} & \frac{\partial \vec{R}}{\partial y} \cdot \frac{\partial \vec{R}}{\partial x} \\ \frac{\partial \vec{R}}{\partial x} \cdot \frac{\partial \vec{R}}{\partial y} & \frac{\partial \vec{R}}{\partial y} \cdot \frac{\partial \vec{R}}{\partial y} \end{bmatrix} \qquad g_{p^i} = \begin{bmatrix} \frac{\partial \vec{R}}{\partial r} \cdot \frac{\partial \vec{R}}{\partial r} & \frac{\partial \vec{R}}{\partial \theta} \cdot \frac{\partial \vec{R}}{\partial r} \\ \frac{\partial \vec{R}}{\partial r} \cdot \frac{\partial \vec{R}}{\partial \theta} & \frac{\partial \vec{R}}{\partial \theta} \cdot \frac{\partial \vec{R}}{\partial \theta} \end{bmatrix}
$$

Allright, now using the Einstein notation to save some writing:

$$
\begin{aligned}
\left\| \frac{d\vec{R}}{d\lambda} \right\|^2 &= \frac{d\vec{R}}{d\lambda} \cdot \frac{d\vec{R}}{d\lambda} \\
&= \frac{dc^i}{d\lambda} \frac{dc^j}{d\lambda} \left(\frac{\partial \vec{R}}{\partial c^i} \cdot \frac{\partial \vec{R}}{\partial c^j} \right) = \frac{dc^i}{d\lambda} \frac{dc^j}{d\lambda} g_{ij} \\
&= \frac{dp^i}{d\lambda} \frac{dp^j}{d\lambda} \left(\frac{\partial \vec{R}}{\partial p^i} \cdot \frac{\partial \vec{R}}{\partial p^j} \right) = \frac{dp^i}{d\lambda} \frac{dp^j}{d\lambda} \tilde{g_{ij}}
\end{aligned}
$$

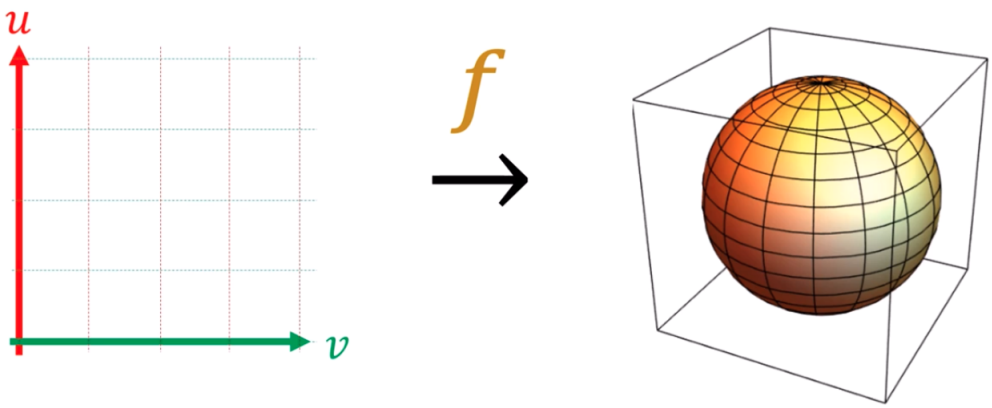
And its transformation rules:

$$
\begin{aligned}
\tilde{g_{ij}} &= \frac{\partial c^k}{\partial p^i} \frac{\partial c^l}{\partial p^j} g_{kl} \\
g_{kl} &= \frac{\partial p^i}{\partial c^k} \frac{\partial p^j}{\partial c^l} \tilde{g_{ij}}
\end{aligned}
$$

6.2 Arc length in curved spaces

Now we'll move on computing arc lengths for curved space, and in particular we're going to focus on the surface of sphere for our maine example.

The first question we can ask ourselves is, how do we describe a 2D surface mathematically? Well, the most common and straight forward way is to start with a flat 2D plane, with u and v coordinates, and then put this into some sort of function f which will stretch and transform the plane so that it becomes a 2D surface living in the 3D space basically:



The variables u and v become parameters of this vector function f (i.e. its **domain** D contained in $\mathbb{R}^2$). The range of the surfaces is contained in the three dimensional space $\mathbb{R}^3$, thus, a surface x is a mapping of D into $\mathbb{R}^3$.
This can be denoted by:

$$f : D \to \mathbb{R}^3 \tag{30}$$

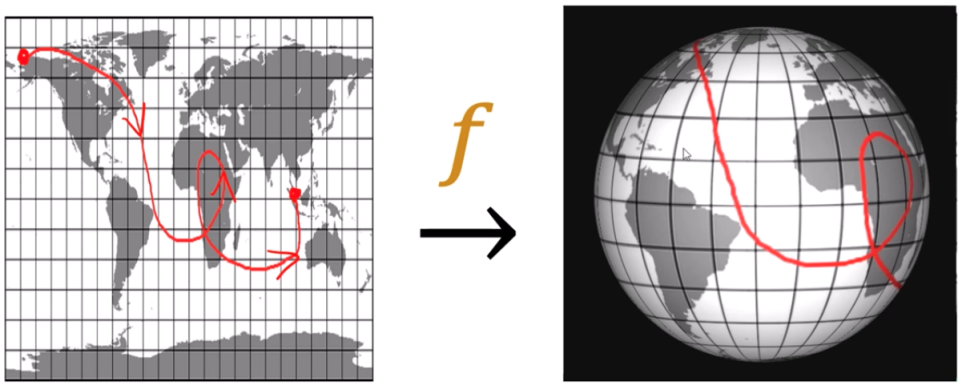
and also represented in 3D coordinate system:

$$(u, v) \to (X(u, v), Y(u, v), Z(u, v))$$

For a sphere surface, its parametrization/transformation function can be written as:

$$\begin{aligned} X &= \cos(v)\sin(u) \\ Y &= \sin(v)\sin(u) \\ Z &= \cos(u) \end{aligned}$$

Allright, and how do we represent a curve on a surface? Well we can apply the same logic, and make the curve, described by γ with parameter t on the uv-plane go through the same transformation function f, to get the curve in the 3D space:



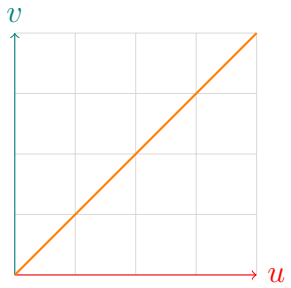
A more mathematically correct definition might be, that given a parametrization for a surface and a curve γ, the curve is on the surface, if a parametrization of γ with a parameter t can be obtained from equations of the surface when u and v become functions of t:

$$\gamma(t) = f(u(t), v(t)) \tag{31}$$

Let's see a simple example of a curve given by:

$$\lambda \to (u = \lambda, v = \lambda) \tag{32}$$

which is just a simple diagonal line:



Its parametrization on the spherical surface can be obtained by applying the transformation functions we saw:

$$\begin{aligned} X &= \cos(\lambda)\sin(\lambda) \\ Y &= \sin(\lambda)\sin(\lambda) \\ Z &= \cos(\lambda) \end{aligned}$$

If we now want to compute the length of such curve, we need to compute the magnitude of the tangent vector $\frac{dR}{d\lambda}$ where R is just the position vector of a point on the curve, in the 3D space X,Y,Z.

If we try to do that, we obtain:

$$
\begin{aligned}
\left\| \frac{d\vec{R}}{d\lambda} \right\|^2 &= \frac{d\vec{R}}{d\lambda} \cdot \frac{d\vec{R}}{d\lambda} \\
&= \left(\frac{\partial \vec{R}}{\partial X}\frac{dX}{d\lambda} + \frac{\partial \vec{R}}{\partial Y}\frac{dY}{d\lambda} + \frac{\partial \vec{R}}{\partial Z}\frac{dZ}{d\lambda} \right) \cdot \left(\frac{\partial \vec{R}}{\partial X}\frac{dX}{d\lambda} + \frac{\partial \vec{R}}{\partial Y}\frac{dY}{d\lambda} + \frac{\partial \vec{R}}{\partial Z}\frac{dZ}{d\lambda} \right) \\
&= \left(\frac{dX}{d\lambda} \right)^2 \left(\frac{\partial \vec{R}}{\partial X} \cdot \frac{\partial \vec{R}}{\partial X} \right) + \frac{dX}{d\lambda}\frac{dY}{d\lambda} \left(\frac{\partial \vec{R}}{\partial X} \cdot \frac{\partial \vec{R}}{\partial Y} \right) + \frac{dX}{d\lambda}\frac{dZ}{d\lambda} \left(\frac{\partial \vec{R}}{\partial X} \cdot \frac{\partial \vec{R}}{\partial Z} \right) \\
&+ \frac{dY}{d\lambda}\frac{dX}{d\lambda} \left(\frac{\partial \vec{R}}{\partial Y} \cdot \frac{\partial \vec{R}}{\partial X} \right) + \left(\frac{dY}{d\lambda} \right)^2 \left(\frac{\partial \vec{R}}{\partial Y} \cdot \frac{\partial \vec{R}}{\partial Y} \right) + \frac{dY}{d\lambda}\frac{dZ}{d\lambda} \left(\frac{\partial \vec{R}}{\partial Y} \cdot \frac{\partial \vec{R}}{\partial Z} \right) \\
&+ \frac{dZ}{d\lambda}\frac{dX}{d\lambda} \left(\frac{\partial \vec{R}}{\partial Z} \cdot \frac{\partial \vec{R}}{\partial X} \right) + \frac{dZ}{d\lambda}\frac{dY}{d\lambda} \left(\frac{\partial \vec{R}}{\partial Z} \cdot \frac{\partial \vec{R}}{\partial Y} \right) + \left(\frac{dZ}{d\lambda} \right)^2 \left(\frac{\partial \vec{R}}{\partial Z} \cdot \frac{\partial \vec{R}}{\partial Z} \right)
\end{aligned}
$$

Due to the orthonormality of the cartesian 3D basis vectors, you would end up with the simple Pythagora's theorem in 3D in this specific case:

$$
\begin{aligned}
\left\| \frac{d\vec{R}}{d\lambda} \right\|^2 &= \frac{d\vec{R}}{d\lambda} \cdot \frac{d\vec{R}}{d\lambda} \\
&= \left(\frac{dX}{d\lambda} \right)^2 + \left(\frac{dY}{d\lambda} \right)^2 + \left(\frac{dZ}{d\lambda} \right)^2
\end{aligned}
$$

So we can now proceed computing these derivatives of X, Y, Z wrt λ and get:

$$
\begin{aligned}
\frac{dX}{d\lambda} &= \cos(2\lambda) \\
\frac{dY}{d\lambda} &= \sin(2\lambda) \\
\frac{dZ}{d\lambda} &= -\sin(\lambda)
\end{aligned}
$$

$$
\begin{aligned}
\left\| \frac{d\vec{R}}{d\lambda} \right\|^2 &= \frac{d\vec{R}}{d\lambda} \cdot \frac{d\vec{R}}{d\lambda} \\
&= \left(\frac{dX}{d\lambda} \right)^2 + \left(\frac{dY}{d\lambda} \right)^2 + \left(\frac{dZ}{d\lambda} \right)^2 \\
&= \left(\cos(2\lambda)^2 \right) + \left(\sin(2\lambda)^2 \right) + \left(-\sin(\lambda)^2 \right) \\
&= 1 + \sin(\lambda)^2
\end{aligned}
$$

So finally, our arc length is:

$$\begin{aligned}\text{arc length} &= \int_0^1 \left\| \frac{d\vec{R}}{d\lambda} \right\| \\ &= \int_0^1 \sqrt{1+\sin(\lambda)^2}\, d\lambda \approx 1.12389\end{aligned}$$

It's interesting to notice that this is not equal to the length of the same curve in the flat uv-plane, which would just be $\sqrt{2}$.
That big ugly formula we used in the first place, can be rewritten as a series of row array (1 x 3) - matrix (3 x 3) - column array (3 x 1) multiplications.

The 3 x 3 matrix composed by the dot products of the basis vectors, is nothing but the metric tensor in the 3D cartesian coordinate system, which we know actually being indeed the 3 x 3 Identity matrix at the end of the day, due to the orthonormality of the basis.

Something we need to realize, is that to compute this arc length in the curved surface (sphere in this case) we used the XYZ coordinate system, so even though we were analyzing the 2D surface, we then worked in the 3D space around the surface, in order to understand it. This approach of studying surfaces is called **extrinsic** geometry, because it involves analyzing curved 2D spaces, using the sorrounding 3D space that the surface lives in.
The other way of analyzing surfaces is without looking at the sorrounding space, where we would only be looking at the uv-plane, and this point of view is called **intrinsic** geometry.

Using the instrinsic approach, instead of studying the tangent vectors in the XYZ space, we would be studying it in terms of the u and v 2D coordinates:

$$\frac{d\vec{R}}{d\lambda} = \left(\frac{\partial \vec{R}}{\partial u}\frac{du}{d\lambda} + \frac{\partial \vec{R}}{\partial v}\frac{dv}{d\lambda} \right) \tag{33}$$

Calculating the magnitude using this tangent vector would give us, in a more compact Einstein notation, considering $u, v = u^1, u^2$:

$$\begin{aligned}\left\| \frac{d\vec{R}}{d\lambda} \right\|^2 &= \frac{d\vec{R}}{d\lambda} \cdot \frac{d\vec{R}}{d\lambda} \\ &= \left(\frac{\partial \vec{R}}{\partial u^i}\frac{du^i}{d\lambda} \right) \cdot \left(\frac{\partial \vec{R}}{\partial u^j}\frac{du^j}{d\lambda} \right) \\ &= \frac{du^i}{d\lambda}\frac{du^j}{d\lambda} \left(\frac{\partial \vec{R}}{\partial u^i} \cdot \frac{\partial \vec{R}}{\partial u^j} \right) \\ &= \frac{du^i}{d\lambda}\frac{du^j}{d\lambda} g_{ij}\end{aligned}$$

where g_{ij} is the metric tensor of the 2D uv-plane, i.e. the dot products of the basis vectors.

Allright, and how do we get the tangent space basis for the uv-plane? So we can compute the metric tensor...well you remember that:

$$\begin{aligned} X &= \cos(v)\sin(u) \\ Y &= \sin(v)\sin(u) \\ Z &= \cos(u) \end{aligned}$$

We can expand the tangent basis vectors using the chain rule:

$$\begin{aligned} \frac{\partial \vec{R}}{\partial u} &= \frac{\partial X}{\partial u}\frac{\partial \vec{R}}{\partial X} + \frac{\partial Y}{\partial u}\frac{\partial \vec{R}}{\partial Y} + \frac{\partial Z}{\partial u}\frac{\partial \vec{R}}{\partial Z} \\ \frac{\partial \vec{R}}{\partial v} &= \frac{\partial X}{\partial v}\frac{\partial \vec{R}}{\partial X} + \frac{\partial Y}{\partial v}\frac{\partial \vec{R}}{\partial Y} + \frac{\partial Z}{\partial v}\frac{\partial \vec{R}}{\partial Z} \end{aligned}$$

Calculating those partial derivatives we get the tangent basis vectors $\vec{e_u}, \vec{e_v}$ in the uv-plane, as linear combination of the 3D basis vectors X, Y, Z:

$$\begin{aligned} \vec{e_u} &= \cos(v)\cos(u)\vec{e_x} + \sin(v)\cos(u)\vec{e_y} - \sin(u)\vec{e_z} \\ \vec{e_v} &= -\sin(v)\sin(u)\vec{e_x} + \cos(v)\sin(u)\vec{e_y} \end{aligned}$$

And now we're able to compute the metric tensor using the basis vector dot products, and performing the calculations you will end up with:

$$\begin{bmatrix} \vec{e_u}\cdot\vec{e_u} & \vec{e_u}\cdot\vec{e_v} \\ \vec{e_v}\cdot\vec{e_u} & \vec{e_v}\cdot\vec{e_v} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \sin(u)^2 \end{bmatrix} \tag{34}$$

And finally, we can compute the arc length as per the equation above:

$$\begin{aligned} \left\| \frac{d\vec{R}}{d\lambda} \right\|^2 &= \frac{d\vec{R}}{d\lambda}\cdot\frac{d\vec{R}}{d\lambda} \\ &= \left(\frac{\partial \vec{R}}{\partial u^i}\frac{du^i}{d\lambda}\right)\cdot\left(\frac{\partial \vec{R}}{\partial u^j}\frac{du^j}{d\lambda}\right) \\ &= \frac{du^i}{d\lambda}\frac{du^j}{d\lambda}\left(\frac{\partial \vec{R}}{\partial u^i}\cdot\frac{\partial \vec{R}}{\partial u^j}\right) \\ &= \frac{du^i}{d\lambda}\frac{du^j}{d\lambda}g_{ij} \\ &= \begin{bmatrix} \frac{du}{d\lambda} & \frac{dv}{d\lambda} \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & \sin(u)^2 \end{bmatrix}\begin{bmatrix} \frac{du}{d\lambda} \\ \frac{dv}{d\lambda} \end{bmatrix} \\ &= \left(\frac{du}{d\lambda}\right)^2 + \sin(u)^2\left(\frac{dv}{d\lambda}\right)^2 \end{aligned}$$

If the arc length we're interested about is given, as we said before, by simple:

$$\lambda \to (u = \lambda, v = \lambda) \tag{35}$$

Then we have that $\frac{du}{d\lambda} = 1$ and $\frac{dv}{d\lambda} = 1$, so we end up with exactly the same result we got when using the extrinsic approach:

$$\begin{aligned} \text{arc length} &= \int_0^1 \left\| \frac{d\vec{R}}{d\lambda} \right\| \\ &= \int_0^1 \sqrt{1 + \sin(\lambda)^2} d\lambda \approx 1.12389 \end{aligned}$$

Main take away from this is that, in curved space, you can't trust your eyeballs, you can only trust the metric tensor

You may wonder one thing, based on the above it seems that in order to get the intrinsic metric tensor matrix, you need to reference the "outside space" anyway. That's actually somewhat true, but there are options to get the intrinsic one from several places:

- Using indeed the extrinsic basis
- Invent one out of imagination
- Obtain it from another equation, like the Einstein Field Equations

Once you have the intrinsic matric, we then can do ALL calculations involving distances and angles, completely intrinsically. But you need that first, and sometimes this involves getting help from the outside space at the start.

The last thing that may still bother you, is the fact that, based on our previous discussions, even in intrinsic geometry we refer to the position vector $\vec{R}$, which references the space outside our curved surface.
If we want to stay completely intrinsic in the surface, we can just remove the position vector from the equations completely, and treat the derivative operators themselves as the vectors

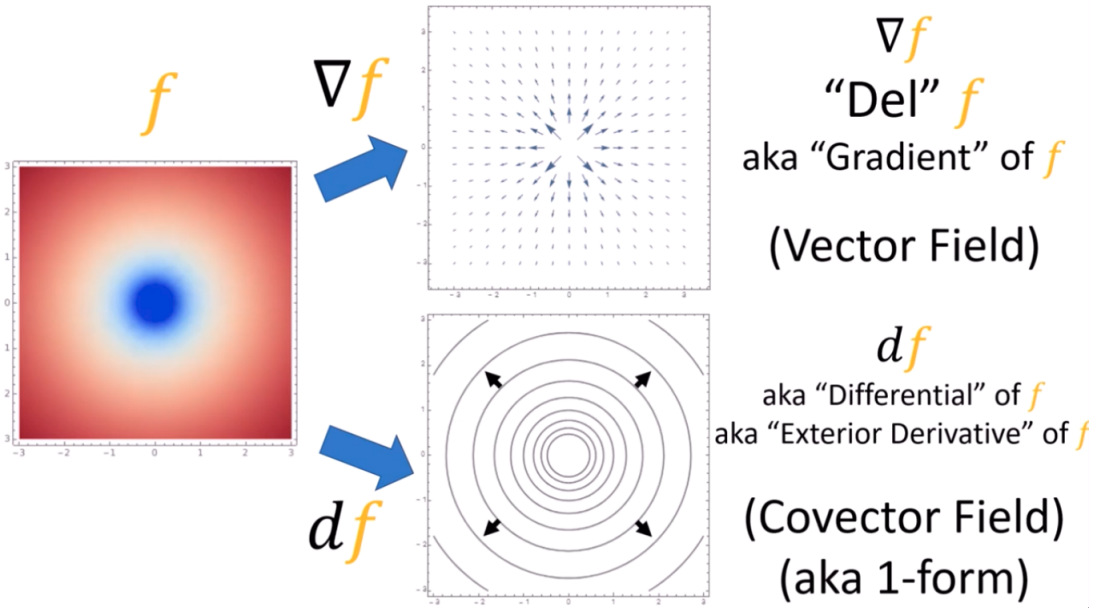
7 Gradient vs d-operator

This sections explains the difference between these two operators, which many times can be confused. And even though they're technically different things, they're highly related to each other.

First of all, we're going to call the operator:

- $\boldsymbol{\nabla} f$ the gradient of the scalar field f
- df the differential of f, also known as the exterior derivative of f

Both the gradient and the differential, act on a scalar field.
The gradient of the scalar field gives us a vector field, where the vectors point in the direction of steepest increase, and larger vectors indicating a steeper change, whereas df gives us a covector field, (also called 1-form), where the covector curves are given by the level sets of the scalar field. And the level sets are oriented towards the positive values of the field.
So it's clear that the two are different, but at the same time fairly similar and highly related.



We know that there's a special connection between a vector space and its dual, and this special connection is given by pairing up a vector $\vec{v}$ with its dual partner covector $\vec{v}\cdot\ -$ and we saw this in our notes on raising/lowering indexes.
The connection between the two partners is possible via the metric tensor.
A similar pairing is actually possible between $\boldsymbol{\nabla} f$ and df.

Let's recall that $df(\vec{v})$ is the directional derivative of f in direction $\vec{v}$ and we can write:

$$df(\vec{v}) = \boldsymbol{\nabla}_{\vec{v}} f = \boldsymbol{\nabla} f \cdot \vec{v} \tag{36}$$

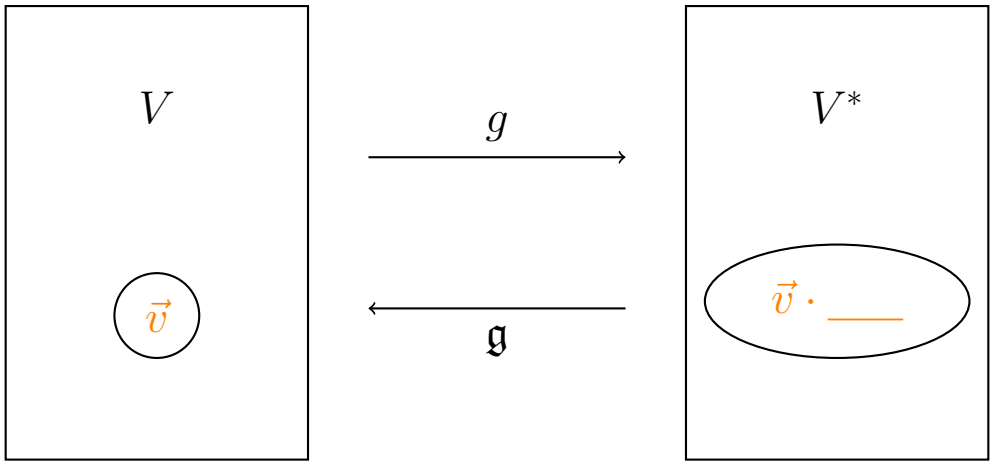
This is already interesting, on the left we have the differential acting on a vector, and on the right we have the dot product of the gradient per the same vector.

In a sense, this means that $df = \boldsymbol{\nabla} f \cdot\ -$.

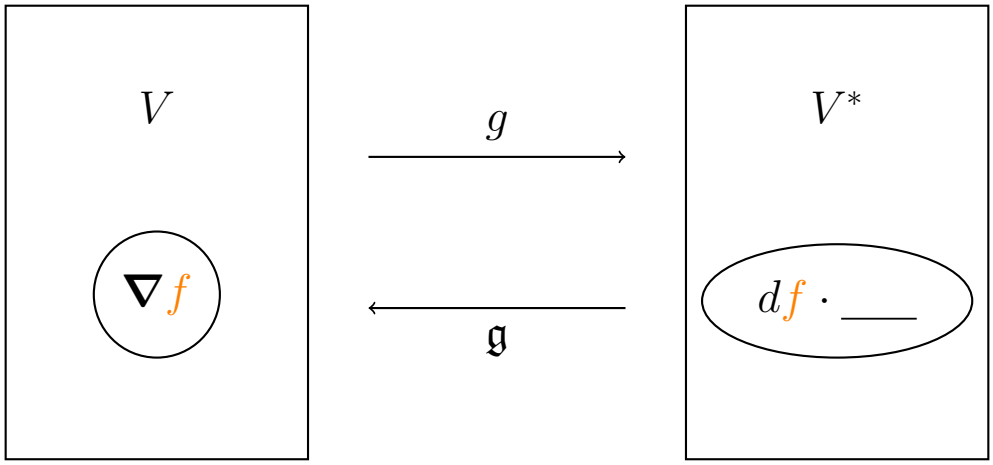
Now, remember that from the raising/lowering indexes notes, we know that:

$$\vec{v}\cdot\ -\ = v^i g_{ij} \epsilon^j \tag{37}$$

And:



As this is true for vector and dual vector (vector-dot-something), the same is true between the gradient and the d-operator:



We can prove this writing down the component form of the operation of df on a vector:

$$\begin{aligned} df(\vec{v}) = \boldsymbol{\nabla} f \cdot \vec{v} &= \left((\boldsymbol{\nabla} f)^i \, \vec{e_i} \right) \cdot \left(v^j \vec{e_j} \right) \\ &= (\boldsymbol{\nabla} f)^i \, v^j \, (\vec{e_i} \cdot \vec{e_j}) \\ &= (\boldsymbol{\nabla} f)^i \, v^j \, g_{ij} \end{aligned}$$

This is the formula for the directional derivative.
But notice here, that we can perform the same approach we did when lowering/raising indexes, so basically taking out the vector as it appears on both sides, and write down in covector components, the $df \cdot \, - \,$, so using the covector basis dc^j

$$df = \boldsymbol{\nabla} f \cdot \, - \, = (\boldsymbol{\nabla} f)^i \, g_{ij} dc^j \tag{38}$$

From previous notes, we also know that we can expand the covector field in its components in this way:

$$df = \frac{\partial f}{\partial c^i} dc^i \tag{39}$$

Putting these together, we can write:

$$df = \frac{\partial f}{\partial c^j} dc^j = (\boldsymbol{\nabla} f)^i \, g_{ij} dc^j \tag{40}$$

Since the covector basis it the same, the components must be equal:

$$\frac{\partial f}{\partial c^j} = (\boldsymbol{\nabla} f)^i \, g_{ij} \tag{41}$$

And the same is true in the other direction, but using the inverse metric tensor:

$$(\boldsymbol{\nabla} f)^k = \mathfrak{g}^{jk} \frac{\partial f}{\partial c^j} \tag{42}$$

This last formula here is very interesting, because it gives us the true formula for the components of the gradient, even though you may already have done that (calculating the gradient of a scalar vector) but never realized that there's the inverse metric tensor involved.

We can write the gradient as a linear combination of the basis vectors like:

$$\boldsymbol{\nabla} f = (\boldsymbol{\nabla} f)^k \frac{\partial}{\partial c^k} = \mathfrak{g}^{jk} \frac{\partial f}{\partial c^j} \frac{\partial}{\partial c^k}$$

This is easily visible as soon as we move from one basis to another, and you can know see the real power of the metric tensor and its inverse, when having to calculate the gradient in different basis.

No strange things happen in cartesian basis, as the metric tensor is the simple identity matrix, hence the inverse is also the identity matrix, so the gradient itself is simply $\boldsymbol{\nabla} f = \frac{\partial f}{\partial c^i} \vec{e_i}$, with the same components as $df = \frac{\partial f}{\partial c^i} dc^i$

The same does not happen in polar or spherical coordinates though, and as you may already know, the gradient components are requiring a scaling factor, which is indeed given by the components of the inverse metric tensor in the specific basis.

In polar coordinates for example, we know that the metric tensor and its inverse are:

$$\tilde{g_{ij}} = \begin{bmatrix} 1 & 0 \\ 0 & r^2 \end{bmatrix} \qquad\qquad \tilde{\mathfrak{g}^{ij}} = \begin{bmatrix} 1 & 0 \\ 0 & 1/r^2 \end{bmatrix}$$

So, the components of the gradient in this system are:

$$\begin{aligned} \boldsymbol{\nabla} f &= \tilde{\mathfrak{g}^{jk}} \frac{\partial f}{\partial p^j} \frac{\partial}{\partial p^k} \\ &= \tilde{\mathfrak{g}^{rr}} \frac{\partial f}{\partial r} \frac{\partial}{\partial r} + \tilde{\mathfrak{g}^{\theta\theta}} \frac{\partial f}{\partial \theta} \frac{\partial}{\partial \theta} \\ &= \frac{\partial f}{\partial r} \frac{\partial}{\partial r} + \frac{1}{r^2} \frac{\partial f}{\partial \theta} \frac{\partial}{\partial \theta} \end{aligned}$$

This might look different from the polar gradient formula you may know from calculus books, $\boldsymbol{\nabla} f = \frac{\partial f}{\partial r}\hat{e}_r + \frac{1}{r}\frac{\partial f}{\partial \theta}\hat{e}_\theta$. The difference is only the basis: textbooks use the *unit* vectors $\hat{e}_r, \hat{e}_\theta$, while here we use the coordinate basis $\frac{\partial}{\partial r}, \frac{\partial}{\partial \theta}$. Since $\frac{\partial}{\partial \theta} = r\,\hat{e}_\theta$ (it has length r), the second term becomes $\frac{1}{r^2}\frac{\partial f}{\partial \theta}\, r\,\hat{e}_\theta = \frac{1}{r}\frac{\partial f}{\partial \theta}\hat{e}_\theta$, matching the textbook result.

To give a bit more formal proof, it's good to recall the tensor product of two covectors, which is a sort of double covector, getting as input two vectors and outputting a scalar, in this way:

$$\begin{aligned}\left(\epsilon^1 \otimes \epsilon^2\right)(\vec{v}, \vec{w}) \\ = \epsilon^1(\vec{v})\epsilon^2(\vec{w})\end{aligned}$$

And remember that we can write them in matrix format as:

$$\begin{aligned}\epsilon^1 \otimes \epsilon^1 &= \begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix} \\ \epsilon^1 \otimes \epsilon^2 &= \begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix} \\ \epsilon^2 \otimes \epsilon^1 &= \begin{bmatrix}0 & 0\\ 1 & 0\end{bmatrix} \\ \epsilon^2 \otimes \epsilon^2 &= \begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix}\end{aligned}$$

So the above tensor product acting on vectors, in matrix/array notation will be:

$$\begin{bmatrix}v^1 & v^2\end{bmatrix}\begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}\begin{bmatrix}w^1\\ w^2\end{bmatrix} = v^1 w^2 \tag{43}$$

If we remember the metric tensor acting on two vectors, this can be written in Einstein notation and then matrix/array notation as:

$$\begin{aligned}g(\vec{v}, \vec{w}) &= v^i w^j g_{ij} \\ &= \begin{bmatrix}v^1 & v^2\end{bmatrix}\begin{bmatrix}g_{11} & g_{12}\\ g_{21} & g_{22}\end{bmatrix}\begin{bmatrix}w^1\\ w^2\end{bmatrix}\end{aligned}$$

So in a sense, we can write down our metric tensor as a linear combination of:

$$\begin{aligned}\begin{bmatrix}g_{11} & g_{12}\\ g_{21} & g_{22}\end{bmatrix} &= g_{11}\begin{bmatrix}1 & 0\\ 0 & 0\end{bmatrix} + g_{12}\begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix} + g_{21}\begin{bmatrix}0 & 0\\ 1 & 0\end{bmatrix} + g_{22}\begin{bmatrix}0 & 0\\ 0 & 1\end{bmatrix} \\ &= g_{ij}(\epsilon^i \otimes \epsilon^j)\end{aligned}$$

We can try to re-proof the below formula in these terms:

$$\vec{v} \cdot\ -\ = v^i g_{ij} \epsilon^j \tag{44}$$

$$
\begin{aligned}
\vec{v} \cdot - &= g(\vec{v}, -) \\
&= \left(g_{ij}(\epsilon^i \otimes \epsilon^j)\right)(v^k e_k) \\
&= g_{ij}\epsilon^i(v^k e_k)\epsilon^j \\
&= g_{ij}v^k\epsilon^i(e_k)\epsilon^j \\
&= g_{ij}v^k\delta^i_k\epsilon^j \\
&= g_{ij}v^i\epsilon^j
\end{aligned}
$$

Here in Tensor Calculus, we can replace the basis vectors/covectors with basis vector/covector fields, hence we can write:

$$g = g_{ij}(dc^i \otimes dc^j) \tag{45}$$

Where these are the basis covector fields in cartesian coordinates ($c^1 = x$, $c^2 = y$), and obeying:

$$dc^j(\frac{\partial}{\partial c^i}) = \delta^j_i \tag{46}$$

And then we can easily proof equation 38

8 Geodesics and Christoffel Symbols in Extrinsic Geometry

In these notes we'll cover Geodesics and Christoffel Symbols in extrinsic geometry, so we're going to consider 2D surfaces living in a 3D space, and we'll cover the same using intrinsic geometry later.

This is also the first step to understanding the concept of Covariant Derivative, but we're getting ahead of ourselves...

Let's start by defining what a geodesic is, a geodesic is the answer to the question of finding the shortest distance between two points.

In a flat space, this is pretty easy, the shortest distance between two points is a straight line connecting the two, in mathematical formulation, we can write, considering two points defined by vectors $\vec{a}$ and $\vec{b}$:

$$\vec{a} + \lambda(\vec{b} - \vec{a}) \tag{47}$$

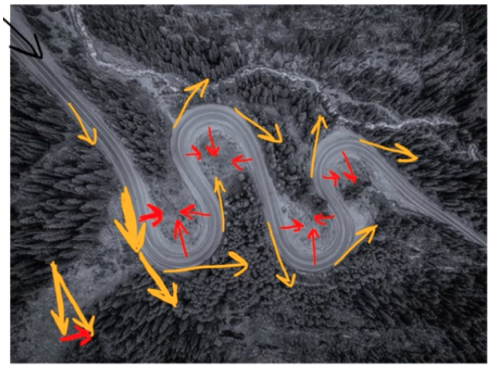
Different values of λ gets us any point on the same line.

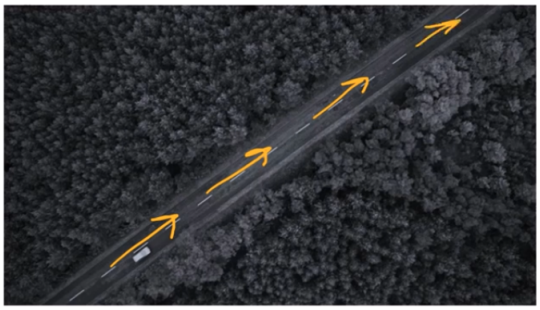
But we generally cannot draw straight lines on a curved 2D surface, so the process of finding the shortest distance between two points involves finding a special curve called the

geodesic, which is basically the straightest possible path in a curved surface/space, that also minimizes the distance between two points.

To start our discussion on geodesics, we can ask ourselves a pretty basic question: how do we know if a path is straight in a flat space? To answer this, we're going to take a look at both velocity and acceleration vectors.

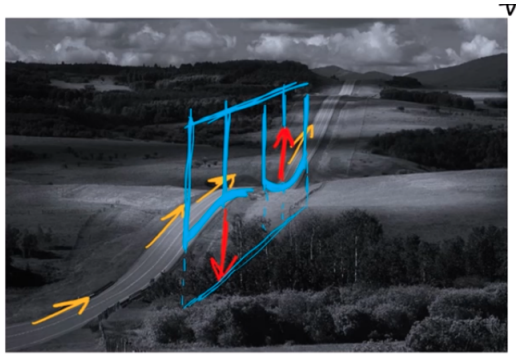
An image talks more than words:





In flat space, a straight path has **zero acceleration** when we travel along it at constant speed.





In curved space, a straight path has **zero tangential acceleration** when we travel along it at constant speed. The acceleration vector is always pointing out of the surface in the normal direction, for a geodesic curve.

And now to the computation of them, we basically need to find curves where the acceleration vector is normal to the surface, so if we consider the extrinsic approach, that would mean that, represending the acceleration vector as composed by two parts, tangential and normal components, the tangential one would need to be zero:

$$\frac{d^2\vec{R}}{d\lambda^2} = \cancel{\left(\frac{d^2\vec{R}}{d\lambda^2}\right)^{tangential}} + \left(\frac{d^2\vec{R}}{d\lambda^2}\right)^{normal}$$

Recall that we've seen how to define a 2D surface using the coordinate system uv and the derivatives of the position vector with respect to these u and v coordinates are the tangent basis vectors $\vec{e_u}, \vec{e_v}$.

So to look at the acceleration vector, we need to compute the second order derivative of the position vector with respect to the parameter λ. We already computed the velocity vector in the tangent plane, in 33, using the multi-variable chain-rule:

$$\frac{d\vec{R}}{d\lambda} = \frac{\partial \vec{R}}{\partial u}\frac{du}{d\lambda} + \frac{\partial \vec{R}}{\partial v}\frac{dv}{d\lambda} \tag{48}$$

So to get the acceleration vector, we compute the derivative:

$$\begin{aligned}
\frac{d}{d\lambda}\left(\frac{d\vec{R}}{d\lambda}\right) &= \frac{d}{d\lambda}\left(\frac{\partial \vec{R}}{\partial u}\frac{du}{d\lambda} + \frac{\partial \vec{R}}{\partial v}\frac{dv}{d\lambda}\right) \\
&= \frac{d}{d\lambda}\left(\frac{du}{d\lambda}\frac{\partial \vec{R}}{\partial u}\right) + \frac{d}{d\lambda}\left(\frac{dv}{d\lambda}\frac{\partial \vec{R}}{\partial v}\right) \\
&= \frac{d^2u}{d\lambda^2}\frac{\partial \vec{R}}{\partial u} + \frac{du}{d\lambda}\left(\frac{d}{d\lambda}\frac{\partial \vec{R}}{\partial u}\right) + \frac{d^2v}{d\lambda^2}\frac{\partial \vec{R}}{\partial v} + \frac{dv}{d\lambda}\left(\frac{d}{d\lambda}\frac{\partial \vec{R}}{\partial v}\right)
\end{aligned}$$

The terms containing the second order derivatives of u and v are okay to understand, but the other two terms, where we see the derivative of the basis vectors with respect to the parameter λ require some additional work to figure out correctly.
We know from multi-variable chain rule, that the λ derivative operator, can be explanded out as a linear combination of u and v derivative operators:

$$\frac{d}{d\lambda} = \frac{du}{d\lambda}\frac{\partial}{\partial u} + \frac{dv}{d\lambda}\frac{\partial}{\partial v} \tag{49}$$

So, the two derivatives above can be expressed as:

$$\begin{aligned}
\frac{d}{d\lambda}\frac{\partial \vec{R}}{\partial u} &= \left(\frac{du}{d\lambda}\frac{\partial}{\partial u} + \frac{dv}{d\lambda}\frac{\partial}{\partial v}\right)\frac{\partial \vec{R}}{\partial u} \\
&= \frac{du}{d\lambda}\frac{\partial^2 \vec{R}}{\partial u^2} + \frac{dv}{d\lambda}\frac{\partial^2 \vec{R}}{\partial u \partial v}
\end{aligned}$$

$$\begin{aligned}
\frac{d}{d\lambda}\frac{\partial \vec{R}}{\partial v} &= \left(\frac{du}{d\lambda}\frac{\partial}{\partial u} + \frac{dv}{d\lambda}\frac{\partial}{\partial v}\right)\frac{\partial \vec{R}}{\partial v} \\
&= \frac{du}{d\lambda}\frac{\partial^2 \vec{R}}{\partial v \partial u} + \frac{dv}{d\lambda}\frac{\partial^2 \vec{R}}{\partial v^2}
\end{aligned}$$

So we can now substitute in the previous expression of the acceleration vector:

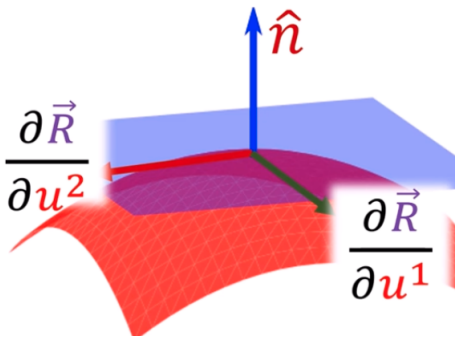
$$\begin{aligned}
\frac{d}{d\lambda}\left(\frac{d\vec{R}}{d\lambda}\right) &= \frac{d^2u}{d\lambda^2}\frac{\partial\vec{R}}{\partial u} + \frac{du}{d\lambda}\left(\frac{d}{d\lambda}\frac{\partial\vec{R}}{\partial u}\right) + \frac{d^2v}{d\lambda^2}\frac{\partial\vec{R}}{\partial v} + \frac{dv}{d\lambda}\left(\frac{d}{d\lambda}\frac{\partial\vec{R}}{\partial v}\right) \\
&= \frac{d^2u}{d\lambda^2}\frac{\partial\vec{R}}{\partial u} + \frac{du}{d\lambda}\left(\frac{du}{d\lambda}\frac{\partial^2\vec{R}}{\partial u^2} + \frac{dv}{d\lambda}\frac{\partial^2\vec{R}}{\partial u\partial v}\right) + \frac{d^2v}{d\lambda^2}\frac{\partial\vec{R}}{\partial v} + \frac{dv}{d\lambda}\left(\frac{du}{d\lambda}\frac{\partial^2\vec{R}}{\partial v\partial u} + \frac{dv}{d\lambda}\frac{\partial^2\vec{R}}{\partial v^2}\right) \\
&= \frac{d^2u}{d\lambda^2}\frac{\partial\vec{R}}{\partial u} + \left(\frac{du}{d\lambda}\right)^2\frac{\partial^2\vec{R}}{\partial u^2} + \frac{du}{d\lambda}\frac{dv}{d\lambda}\frac{\partial^2\vec{R}}{\partial u\partial v} + \frac{d^2v}{d\lambda^2}\frac{\partial\vec{R}}{\partial v} + \frac{dv}{d\lambda}\frac{du}{d\lambda}\frac{\partial^2\vec{R}}{\partial v\partial u} + \left(\frac{dv}{d\lambda}\right)^2\frac{\partial^2\vec{R}}{\partial v^2}
\end{aligned}$$

To get a bit more compact notation, we can define $u^1 = u$ and $u^2 = v$ and using Einstein summation convention:

$$\frac{d^2\vec{R}}{d\lambda^2} = \frac{d^2u^i}{d\lambda^2}\frac{\partial\vec{R}}{\partial u^i} + \frac{du^i}{d\lambda}\frac{du^j}{d\lambda}\frac{\partial^2\vec{R}}{\partial u^i\partial u^j}$$

Now, the first terms (only summed over index i) are just coefficients (second order derivatives) multiplied by the tangent vectors $\frac{\partial\vec{R}}{\partial u^i}$, so are a linear combination of the tangent vectors, hence we're sure they're purely lying on the tangent plane to the surface.
The same is not valid for the other terms (summed both over i and j), as they measure how the tangent vectors themselves change, so it can have both a tangent component, but also a normal component to the tangent plane on the surface.

Let's then study these terms, and we're going to do to by expaning these in some basis. At the end of the day these are 3D vectors and can be expanded, for example, on the X, Y, Z coordinates we've seen, but we prefer to expand these in the tangent plane using the tangent plane basis vectors themselves $\frac{\partial\vec{R}}{\partial u^i}$.



So we can write these as linear combination with some coefficients we define as Γ for the tangent components, and L for the normal one.

$$\begin{aligned}
\frac{\partial^2\vec{R}}{\partial u^i\partial u^j} &= \Gamma^1_{ij}\frac{\partial\vec{R}}{\partial u^1} + \Gamma^2_{ij}\frac{\partial\vec{R}}{\partial u^2} + L_{ij}\hat{n} \\
&= \Gamma^k_{ij}\frac{\partial\vec{R}}{\partial u^k} + L_{ij}\hat{n}
\end{aligned}$$

The Γ coefficients are called **Christoffel symbols** and gives us the tangential components of that second order derivative, whereas the L coefficients are the coefficients of the **second fundamental form**, which gives us the normal components of that.

We can derive the Christoffel symbols by considering that the normal vector is by definition normal to the tangent basis vectors, so it means that their dot product is 0, we can use this fact and perform the dot product on both sides:

$$\begin{aligned}\frac{\partial^2 \vec{R}}{\partial u^i \partial u^j} \cdot \frac{\partial \vec{R}}{\partial u^l} &= \left(\Gamma_{ij}^k \frac{\partial \vec{R}}{\partial u^k} + L_{ij} \hat{n} \right) \cdot \frac{\partial \vec{R}}{\partial u^l} \\ &= \Gamma_{ij}^k \frac{\partial \vec{R}}{\partial u^k} \cdot \frac{\partial \vec{R}}{\partial u^l} \\ &= \Gamma_{ij}^k g_{kl}\end{aligned}$$

To isolate the Christoffel symbols, we need to get rid of the metric tensor on the right side, and we can do so by recalling the defintion of the inverse metric tensor, so:

$$\begin{aligned}\frac{\partial^2 \vec{R}}{\partial u^i \partial u^j} \cdot \frac{\partial \vec{R}}{\partial u^l} \mathfrak{g}^{lm} &= \Gamma_{ij}^k g_{kl} \mathfrak{g}^{lm} \\ &= \Gamma_{ij}^k \delta_k^m \\ &= \Gamma_{ij}^m\end{aligned}$$

So we have:

$$\Gamma_{ij}^m = \frac{\partial^2 \vec{R}}{\partial u^i \partial u^j} \cdot \frac{\partial \vec{R}}{\partial u^l} \mathfrak{g}^{lm} \tag{50}$$

In a similar way we can derive the coefficients of the second fundamental forms, this time with the dot product with the normal vector itself:

$$\begin{aligned}\frac{\partial^2 \vec{R}}{\partial u^i \partial u^j} \cdot \hat{n} &= \left(\Gamma_{ij}^k \frac{\partial \vec{R}}{\partial u^k} + L_{ij} \hat{n} \right) \cdot \hat{n} \\ &= L_{ij} \hat{n} \cdot \hat{n} \\ &= L_{ij}\end{aligned}$$

So we have, also considering a more concrete expression for the normal vector to the tangent plane of the surface:

$$L_{ij} = \frac{\partial^2 \vec{R}}{\partial u^i \partial u^j} \cdot \frac{\vec{e_1} \times \vec{e_2}}{\|\vec{e_1} \times \vec{e_2}\|} \tag{51}$$

We can finally plug these additional terms into the equation of the second order derivative of the position vector, to end up with:

$$\begin{aligned}\frac{d^2\vec{R}}{d\lambda^2} &= \frac{d^2u^i}{d\lambda^2}\frac{\partial\vec{R}}{\partial u^i} + \frac{du^i}{d\lambda}\frac{du^j}{d\lambda}\frac{\partial^2\vec{R}}{\partial u^i\partial u^j} \\ &= \frac{d^2u^k}{d\lambda^2}\frac{\partial\vec{R}}{\partial u^k} + \frac{du^i}{d\lambda}\frac{du^j}{d\lambda}\left(\Gamma^k_{ij}\frac{\partial\vec{R}}{\partial u^k} + L_{ij}\hat{n}\right)\end{aligned}$$

If you notice, in this equation, all the terms are either given using the tangent vectors, or the normal vector, so we can group the terms together:

$$\frac{d^2\vec{R}}{d\lambda^2} = \left(\frac{d^2u^k}{d\lambda^2} + \Gamma^k_{ij}\frac{du^i}{d\lambda}\frac{du^j}{d\lambda}\right)\frac{\partial\vec{R}}{\partial u^k} + L_{ij}\frac{du^i}{d\lambda}\frac{du^j}{d\lambda}\hat{n}$$

Our main objective was to find geodesic curves, which are curves where the acceleration vector is only normal to the surface. What this means is that the tangential term of the above equation, in order to satisfy the geodesic requirement, has to be 0:

$$\boxed{\frac{d^2u^k}{d\lambda^2} + \Gamma^k_{ij}\frac{du^i}{d\lambda}\frac{du^j}{d\lambda} = 0} \tag{52}$$

8.1 Geodesic Examples on Plane and Sphere

This section gives examples on the actual computation of geodesics, and the plan to do so is:

- Compute Christoffel symbols

$$\boxed{\Gamma^k_{ij} = \frac{\partial^2\vec{R}}{\partial u^i\partial u^j}\cdot\frac{\partial\vec{R}}{\partial u^l}\mathfrak{g}^{lk}} \tag{53}$$

  which gives us information on how the basis vectors change from point to point on the surface.

- Solve Geodesic equations

$$\boxed{\frac{d^2u^k}{d\lambda^2} + \Gamma^k_{ij}\frac{du^i}{d\lambda}\frac{du^j}{d\lambda} = 0} \tag{54}$$

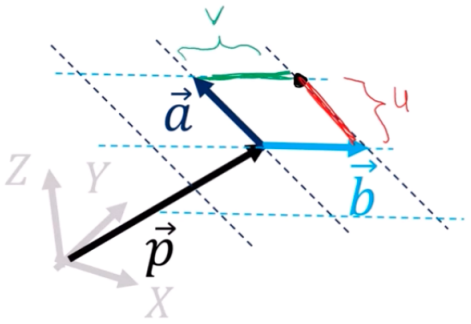
  Plugging the Christoffel symbols and solve for the path, will is a set of u-coordinates in function of the path parameter λ

8.1.1 Geodesics on a plane

Let's start with computing geodesics on a plane surface. This is more a sanity check to actually verify that the geodesic equation is a simple straight line lying on the plane itself, as we already know.

The equation of a 2D plane is given by:

$$\vec{R}(u,v) = \vec{p} + u\vec{a} + v\vec{b} \tag{55}$$



To compute the Christoffel symbols, we need to compute the first and second order derivative of the position vector with respect to the plane's coordinates u and v, which are easily calculated as:

$$\frac{\partial \vec{R}}{\partial u} = \vec{a}, \quad \frac{\partial \vec{R}}{\partial v} = \vec{b}$$
$$\frac{\partial^2 \vec{R}}{\partial u^2} = \frac{\partial^2 \vec{R}}{\partial v^2} = \frac{\partial^2 \vec{R}}{\partial u v} = \frac{\partial^2 \vec{R}}{\partial v u} = \vec{0}$$

Since the Christoffel symbols depend on those second order derivative, and they're all zero, that means that on the 2D plane:

$$\Gamma^k_{ij} = 0 \tag{56}$$

If you think about it, it actually makes sense since Christoffel symbols track how the basis vectors change from point to point, and on a 2D surface, the basis vectors are just constant in every point in space, hence Christoffel symbols are correctly zero.

This makes it easier to compute the geodesic equations, as this only depends on the second-order derivative of the coordinates wrt the parameter:

$$\frac{d^2 u^k}{d\lambda^2} + \cancel{\Gamma^k_{ij} \frac{du^i}{d\lambda} \frac{du^j}{d\lambda}} = 0$$

In other words, geodesics on the plane, are curve for which acceleration is zero.

$$\frac{d^2 u}{d\lambda^2} = 0$$
$$\frac{d^2 v}{d\lambda^2} = 0$$

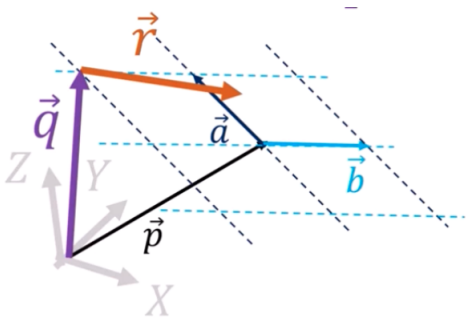
The solution to these differential equations would just be linear equations for both u and v:

$$u = k_u\lambda + u_0$$
$$v = k_v\lambda + v_0$$

where the numbers k_u, u_0, k_v, v_0 are just constants.
Plugging these into the 2D plane equation:

$$\begin{aligned}\vec{R}(u,v) &= \vec{p} + u\vec{a} + v\vec{b} \\ &= \vec{p} + (k_u\lambda + u_0)\,\vec{a} + (k_v\lambda + v_0)\,\vec{b} \\ &= \left(\vec{p} + u_0\vec{a} + v_0\vec{b}\right) + \lambda\left(k_u\vec{a} + k_v\vec{b}\right) \\ &= \vec{q} + \lambda(\vec{r})\end{aligned}$$

All of this, is just a single position vector of a point on the plane ($\vec{q}$), plus a vector ($\vec{r}$) living in the plane, scaled by λ



Where the constants simply specify the initial position (u_0 and v_0) and initial velocity (k_u and k_v).
So, given an initial position and velocity, there's a unique geodesic on the 2D surface, which is just the straight line heading in the direction of the velocity vector.

8.1.2 Geodesics on a sphere

Now we consider a 2D surface of a sphere, of radius 1.
Let's recall the XYZ equations defining the surface, condering the radius=1:

$$X = \cos(v)\sin(u)$$
$$Y = \sin(v)\sin(u)$$
$$Z = \cos(u)$$

We already calculated the first order derivative, i.e. the tangent basis vectors before, using the chain rule, as:

$$\frac{\partial \vec{R}}{\partial u} = \cos(v)\cos(u)\frac{\partial \vec{R}}{\partial X} + \sin(v)\cos(u)\frac{\partial \vec{R}}{\partial Y} - \sin(u)\frac{\partial \vec{R}}{\partial Z}$$
$$\frac{\partial \vec{R}}{\partial v} = -\sin(v)\sin(u)\frac{\partial \vec{R}}{\partial X} + \cos(v)\sin(u)\frac{\partial \vec{R}}{\partial Y}$$

And then the metric tensor:

$$g_{ij} = \begin{bmatrix} \frac{\partial \vec{R}}{\partial u}\cdot\frac{\partial \vec{R}}{\partial u} & \frac{\partial \vec{R}}{\partial u}\cdot\frac{\partial \vec{R}}{\partial v} \\ \frac{\partial \vec{R}}{\partial v}\cdot\frac{\partial \vec{R}}{\partial u} & \frac{\partial \vec{R}}{\partial v}\cdot\frac{\partial \vec{R}}{\partial v} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \sin(u)^2 \end{bmatrix} \tag{57}$$

Proceeding further, to compute the Christoffel symbols we'll need the second-order derivatives, so we can get:

$$\frac{\partial}{\partial u}\left(\frac{\partial \vec{R}}{\partial u}\right) = -\cos(v)\sin(u)\frac{\partial \vec{R}}{\partial X} - \sin(v)\sin(u)\frac{\partial \vec{R}}{\partial Y} - \cos(u)\frac{\partial \vec{R}}{\partial Z}$$
$$\frac{\partial}{\partial v}\left(\frac{\partial \vec{R}}{\partial v}\right) = -\cos(v)\sin(u)\frac{\partial \vec{R}}{\partial X} - \sin(v)\sin(u)\frac{\partial \vec{R}}{\partial Y}$$
$$\frac{\partial}{\partial u}\left(\frac{\partial \vec{R}}{\partial v}\right) = \frac{\partial}{\partial v}\left(\frac{\partial \vec{R}}{\partial u}\right) = -\sin(v)\cos(u)\frac{\partial \vec{R}}{\partial X} + \cos(v)\cos(u)\frac{\partial \vec{R}}{\partial Y}$$

We can also get easily the inverse metric tensor, as the metric tensor is diagonal, so its inverse is formed by the reciprocal of the components:

$$\mathfrak{g}^{ij} = \begin{bmatrix} 1 & 0 \\ 0 & 1/\sin(u)^2 \end{bmatrix} \tag{58}$$

So finally we're ready to compute the Christoffel symbols: it has 3 indexes, i, j and k, and each of those goes from 1 to 2, so that means there are 2 x 2 x 2 symbols to compute, for a total of 8.
Notice also that, in the formula there's also the l index on the right hand side, so there's a summation over l going on as well.

Let's start by considering the k=1 and k=2 equations, and explicitly write the summation over l:

$$\Gamma^1_{ij} = \frac{\partial^2 \vec{R}}{\partial u^i \partial u^j}\cdot\frac{\partial \vec{R}}{\partial u^1}\mathfrak{g}^{11} + \frac{\partial^2 \vec{R}}{\partial u^i \partial u^j}\cdot\frac{\partial \vec{R}}{\partial u^2}\mathfrak{g}^{21}$$
$$\Gamma^2_{ij} = \frac{\partial^2 \vec{R}}{\partial u^i \partial u^j}\cdot\frac{\partial \vec{R}}{\partial u^1}\mathfrak{g}^{12} + \frac{\partial^2 \vec{R}}{\partial u^i \partial u^j}\cdot\frac{\partial \vec{R}}{\partial u^2}\mathfrak{g}^{22}$$

In this way, we clearly see the components of the inverse metric tensor in the equations, and we know that the off-diagonal components are equal to zero, hence, already plugging in the values:

$$\Gamma^1_{ij} = \frac{\partial^2 \vec{R}}{\partial u^i \partial u^j} \cdot \frac{\partial \vec{R}}{\partial u^1} \mathfrak{g}^{11} = \frac{\partial^2 \vec{R}}{\partial u^i \partial u^j} \cdot \frac{\partial \vec{R}}{\partial u^1}$$
$$\Gamma^2_{ij} = \frac{\partial^2 \vec{R}}{\partial u^i \partial u^j} \cdot \frac{\partial \vec{R}}{\partial u^2} \mathfrak{g}^{22} = \frac{\partial^2 \vec{R}}{\partial u^i \partial u^j} \cdot \frac{\partial \vec{R}}{\partial u^2} \frac{1}{\sin(u)^2}$$

And now we need to compute all the dot products.
Normally, the dot product of two 3D vectors, we'd end up with 9 total terms, but since here we're dealing with the cartsian XYZ coordinate system, all the off-diagonal dot products go to zero, and the diagonal terms are simply equal to 1, so at the end, in the cartesian system, the dot product is a simple sum of 3 multiplications:

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$$\vec{a} \cdot \vec{b} = a^X b^X + a^Y b^Y + a^Z b^Z$$

We start with the dot products:

$$\frac{\partial \vec{R}}{\partial u} \cdot \frac{\partial^2 \vec{R}}{\partial u^2} = \begin{bmatrix} \cos(v)\cos(u) \\ \sin(v)\cos(u) \\ -\sin(u) \end{bmatrix} \cdot \begin{bmatrix} -\cos(v)\sin(u) \\ -\sin(v)\sin(u) \\ -\cos(u) \end{bmatrix} = -\cos(u)\sin(u) + \cos(u)\sin(u) = 0$$
$$\frac{\partial \vec{R}}{\partial u} \cdot \frac{\partial^2 \vec{R}}{\partial v^2} = \begin{bmatrix} \cos(v)\cos(u) \\ \sin(v)\cos(u) \\ -\sin(u) \end{bmatrix} \cdot \begin{bmatrix} -\cos(v)\sin(u) \\ -\sin(v)\sin(u) \\ 0 \end{bmatrix} = -\cos(u)\sin(u)$$
$$\frac{\partial \vec{R}}{\partial u} \cdot \frac{\partial^2 \vec{R}}{\partial u \partial v} = \begin{bmatrix} \cos(v)\cos(u) \\ \sin(v)\cos(u) \\ -\sin(u) \end{bmatrix} \cdot \begin{bmatrix} -\sin(v)\cos(u) \\ \cos(v)\cos(u) \\ 0 \end{bmatrix} = 0$$
$$\dots$$
$$\dots$$

You can go ahead with all the calculations and you eventually find that the non-zero Christoffel symbols are:

$$\Gamma^1_{22} = -\cos(u)\sin(u)$$
$$\Gamma^2_{12} = \Gamma^2_{21} = \frac{\cos(u)}{\sin(u)}$$

Note that since the order of differentiation does not matter, you have that $\Gamma^k_{ij} = \Gamma^k_{ji}$

Now, plugging these into the geodesic equations, the only non-zero ones are:

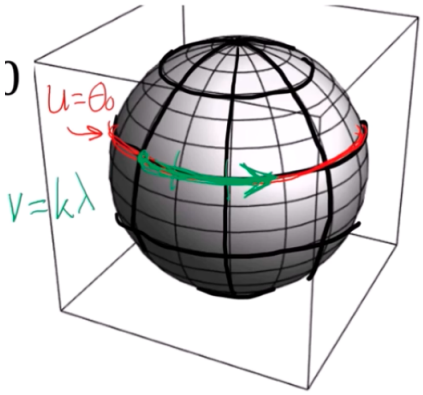
$$\frac{d^2u^1}{d\lambda^2} + \Gamma^1_{22}\frac{du^2}{d\lambda}\frac{du^2}{d\lambda} = 0$$
$$\frac{d^2u^2}{d\lambda^2} + \Gamma^2_{12}\frac{du^1}{d\lambda}\frac{du^2}{d\lambda} + \Gamma^2_{21}\frac{du^2}{d\lambda}\frac{du^1}{d\lambda} = \frac{d^2u^2}{d\lambda^2} + 2\Gamma^2_{12}\frac{du^1}{d\lambda}\frac{du^2}{d\lambda} = 0$$

We can plug in the Christoffel symbols, rewrite the coordinates back to u and v, and replace the $cos/sin = cot = 1/tan$:

$$\frac{d^2u}{d\lambda^2} - \cos(u)\sin(u)\frac{dv}{d\lambda}\frac{dv}{d\lambda} = 0$$
$$\frac{d^2v}{d\lambda^2} + 2\cot(u)\frac{du}{d\lambda}\frac{dv}{d\lambda} = 0$$

Now there are non-linear couple of differential equations, not easy to solve in the general case.
To simplify the example, we can try some special cases like a circle of latitude on the sphere:

$$u = \theta_0$$
$$v = k\lambda$$



In this way we have simple first derivative, and 0 second order derivatives:

$$\frac{du}{d\lambda} = 0$$
$$\frac{dv}{d\lambda} = k$$

$$\frac{d^2u}{d\lambda^2} = 0$$
$$\frac{d^2v}{d\lambda^2} = 0$$

Bringing these into the geodesic equations, we obtain:

$$\cos(\theta_0)\sin(\theta_0)k^2 = 0$$
$$0 = 0$$

The second one is automatically true of course, and to make the first equation true, we need to choose the values of θ_0 and k so that the term goes to 0.

You can set $k = 0$ but that's not very interesting because that's a single point at a constant v. Another option is to make the sin part go to zero by setting $\sin(\theta_0) = 0$, that's basically $\theta_0 = 0$ or $\theta_0 = \pi$, but this also is not very interesting because it's a single point at north/south pole of the sphere.

The other option is to have the cos part go to zero, and this requires $\theta_0 = \pi/2$ which is a curve around the equator.

So the path along the equator

$$u = \pi/2$$
$$v = k\lambda$$

is a geodesic curve on the sphere, which means that the shortest way to get between two points on the equator, is to travel straight along the equator on the geodesic path.
Other circles of latitude are not geodesics, like if you take a latitude path on the northern emisphere, you will find out that the geodesic is not along the same latitude path, but it curves slighlty above it.
The only circle of latitude that it's a geodesic, is the equator, which makes sense because if we think of traveling along the equator, our accelerator vectors always point directly inward towards the center of the circle, which is also the center of the sphere, hence completely normal to the surface (geodesic requirement). Instead, if we travel along another circle of latitude, the accelerator vectors point to the center of the circle, which in this case, is not the center of the sphere, so we end up with a tangential component as well as a normal component.
Now, the equator being a geodesic, by rotational symmetry, all great circles (which center is located at the center of the sphere) are geodesics.

9 Covariant Derivative

If we want to give a one-sentence definition of the Covariant Derivative, we can say it's the **understanding the rate of change of vector (tensor) fields that takes changing basis vectors into account**

There's obviously some explanation and proof required, and we're going to define the Covariant Derivative first in the flat space, which is the easiest way to look at it, and then

follow along, increasing the challenge, by looking at it in curved-space, first extrinsic and then intrinsic, and finally giving its most abstract definition.

9.1 Covariant derivative in flat-space

Let's start by recalling that, given a certain basis, we can define a vector $\vec{v}$ as a linear conbination of the basis vectors, for example in cartesian basis $\vec{e_i}$ or polar $\tilde{\vec{e}}_j$:

$$\vec{v} = v^i \vec{e_i}$$
$$\vec{v} = v^i \tilde{\vec{e}}_i$$

If we try to differentiate the vector with respect to one of the coordinates in cartesian basis in $\mathbb{R}^2$, let's say x:

$$\begin{aligned}
\frac{\partial}{\partial x}(\vec{v}) &= \frac{\partial}{\partial x}\left(v^x \vec{e_x} + v^y \vec{e_y}\right) \\
&= \frac{\partial}{\partial x}\left(v^x \vec{e_x}\right) + \frac{\partial}{\partial x}\left(v^y \vec{e_y}\right) \\
&= \frac{\partial v^x}{\partial x}\vec{e_x} + v^x \cancel{\frac{\partial}{\partial x}(\vec{e_x})} + \frac{\partial v^y}{\partial x}\vec{e_y} + \cancel{v^y \frac{\partial}{\partial x}(\vec{e_y})} \\
&= \frac{\partial v^x}{\partial x}\vec{e_x} + \frac{\partial v^y}{\partial x}\vec{e_y}
\end{aligned}$$

where the canceled components are due to the fact that, in cartesian coordinates, the basis vectors are just constant, so the derivative is reduced to be a linear combination of the basis vectors, with coefficients given by the derivative of the vector components.

Different is the case of polar coordinates though, because the vector basis are not constant in space, but change from point to poing. In polar coordinates, we end up with 4 components, 2 of them track the rate of change of components, and 2 of them track the change of the basis vectors.

In general, this is actually the simplest definition of the covariant derivative, so an ordinary derivative that takes into account the change of basis, so it makes sure to differentiate both the components and the basis vectors.
Writing that down using Einstein summation convention:

$$\begin{aligned}
\frac{\partial}{\partial c^i}(\vec{v}) &= \frac{\partial}{\partial c^i}\left(v^j \vec{e_j}\right) \\
&= \frac{\partial v^j}{\partial c^i}\vec{e_j} + v^j \frac{\partial \vec{e_j}}{\partial c^i}
\end{aligned}$$

Sometimes, the basis vector derivative, is commonly expanded into a linear combination of the basis vectors themselves, so:

$$\frac{\partial \vec{e_j}}{\partial c^i} = \Gamma^1_{ij}\vec{e_1} + \Gamma^2_{ij}\vec{e_2}$$
$$= \Gamma^k_{ij}\vec{e_k}$$

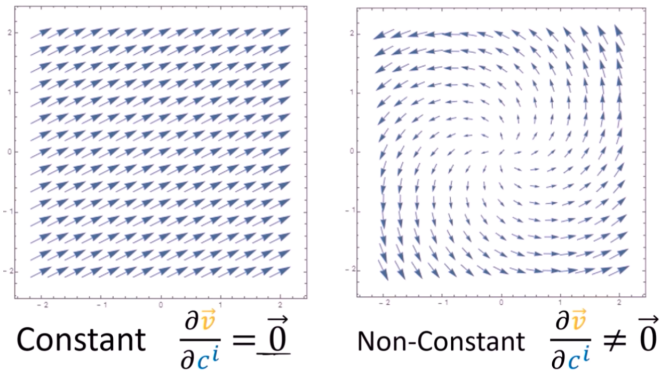
So the covariant derivative can be re-written as:

$$\frac{\partial}{\partial c^i}(\vec{v}) = \frac{\partial v^k}{\partial c^i}\vec{e_k} + v^j\Gamma^k_{ij}\vec{e_k}$$
$$= \left(\frac{\partial v^k}{\partial c^i} + v^j\Gamma^k_{ij}\right)\vec{e_k}$$

9.2 Covariant derivative in curved-space - extrinsic

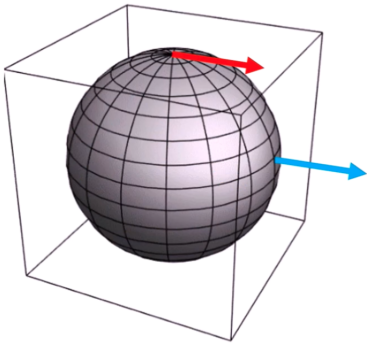
So now we're moving on into looking at the rate of change of vector fields, on curved surfaces. It turns out that it's a lot trickier dealing with vector spaces here, when compared to flat ones, and we're going to explain why.

Let's take these two vector fields here below, you can clearly visualize that the first one is constant, whereas the second one is not:



The first one is constant because we can see that all the vectors are the same...but how do we say that? It may sounds silly and straight forward, but how do we know that these are the same? Well what we could do is take a couple of these vectors shown in the first picture, and slide one on top of the other, and check if they overlap perfectly. That's quite easy to do right?

But let's consider instead, a vector field existing on the surface of a 2D sphere, so every point on the sphere, has a vector associated to it. And now let's try to ask the same question: if we compare the blue vector and the red vector in the picture, how do we tell if they're the same?



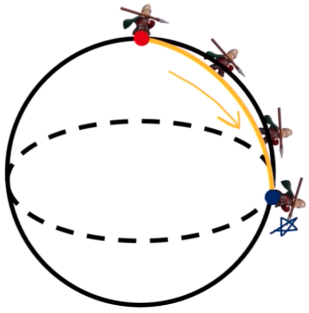
You might be tempted to do exactly the same thing we did before, taking the red vector, slide it over to the red and find out that they overlap perfectly. There's a problem with this method though, and you can maybe realize it already.

If you imagine this sphere being a planet, and you stand still on the north pole where the red vector's origin is. To you, the red vector would look like pointing forward towards the horizon. If you imagine instead, standing still on the blue vector's origin point, the blue vector would look to you, as pointing towards the sky, perpendicular to the surface.
So from the point of view of someone living on the 2D surface, these two vectors are clearly not the same.

That means that if we're going to invent a derivative that makes sense for the "inhabitants" of the 2D surface, we need to come out with a new way of comparing vectors on the surface, that works for the observers living on the surface.

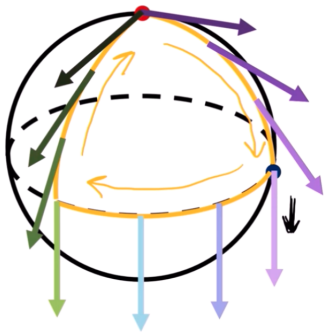
We're going to introduce an idea called **Parallel transport:** a way of moving vectors around on a surface that "keeps them straight".

How does this work? Imagine you standing on the sphere's north pole, holding an arrow towards the horizon in front of you. Now imagine you want to transport this arrow down towards the equator of the sphere, while holding the arrow as straight as possible.
What this means is that while you walk along the surface towards the equator, the arrow will always point towards the horizon from your point of view, so it would look something like this:



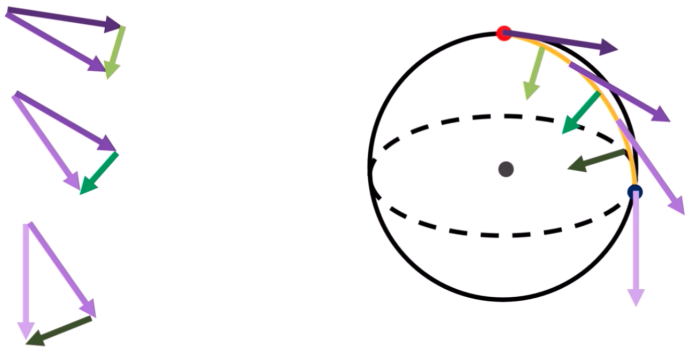
Of course, you can probably already see a problem with this parallel transport: if you imagine now, from the point of view of this person holding the arrow, traveling along the

equator, always holding the arrow straight to him, and then walking backwards to the point it started on the north pole, you will find that the arrow/vector transported does not look like when it started, so the starting and ending vectors are not the same:



This means that **parallel transport DOES NOT keep vectors constant.**
The reason for that is that on a curved surface **parallel transport is path-dependent**: transporting the same vector along two different paths to the same point generally gives two different results, so in general there's no way to define a globally constant vector field.
The parallel transport keeps vectors "as constant as possible" while moving along the surface step-by-step, and this is the best we can do, and the downside is that if traveling in a loop, we may and up with a vector that looks differently than the one with started with.

Allright, now that we have the intuition, how do we describe this mathematically? Referring to the last figure above, if we consider the first two vectors , the first at the north pole, and the next one moving towards the equator, and compare them, we can see that the change between them is this vector below, which points towards the center of the sphere, the same if we continue comparing vectors in pair:



So this means that, when we parallel transport a vector "forward", its rate of change is completely normal to the surface, perpendicular to the tangent plane.
Bringing this vector along this path, basically gives us a vector field defined at every point along this curve, and this vector field obeys this relationship:

$$\frac{d\vec{v}}{d\lambda} = \left(\frac{d\vec{v}}{d\lambda} \cdot \hat{n}\right) \hat{n} \tag{59}$$

i.e. the rate of change is purely normal: it equals its own projection onto the unit normal $\hat{n}$. Or, in other words, the difference between the rate of change of this vector field, and its normal part, is equal to $\vec{0}$:

$$\frac{d\vec{v}}{d\lambda} - \left(\frac{d\vec{v}}{d\lambda} \cdot \hat{n}\right)\hat{n} = \vec{0} \tag{60}$$

So, the best that we can do to define a "constant" vector field on a curved surface, is parallel transporting a vector along a curve, and this, is where the covariant derivative on curved surfaces come in.

The covariant derivative on curved surfaces helps us find parallel transported vector fields.

So we're going to define the covariant derivative on a curved surface, as the rate of change of a vector field $\vec{v}$ in some direction $\vec{w}$, with the normal component, of the rate of change, subtracted..

$$\nabla_{\vec{w}}\vec{v} \tag{61}$$

So let's say we have a curve parametrized by the parameter λ, and as the parameter increases, we move forward along the curve. We have a vector field $\vec{v}$ defined along this curve, and we want to find the rate of change of this vector field as we travel along the curve.

The covariant derivative of this vector field in a direction given by $d/d\lambda$, which is basically the direction along the curve, would be the rate of change of the vector field, in the lambda direction, with the component normal to the surface subtracted.

$$\nabla_{\frac{d}{d\lambda}}\vec{v} = \frac{d\vec{v}}{d\lambda} - \left(\frac{d\vec{v}}{d\lambda} \cdot \hat{n}\right)\hat{n} \tag{62}$$

And as we said, when this difference quantity is zero, it's when the vector field is being parallel transported along the curve.

So we can say that, when the covariant derivative $\nabla_{\vec{w}}\vec{v} = \vec{0}$, it means that the vector field is parallel transported in the direction $\vec{w}$.

All of this can be clarified by giving an example, of the covariant derivative + parallel transport on a sphere of radius = 1.
We'll use the 2D plane of coordinates u^1 and u^2, knowing from previous section, that the parametrization of the sphere in 3D, using these 2D coordinates can be written as:

$$\begin{aligned} X &= \cos\left(u^2\right)\sin\left(u^1\right) \\ Y &= \sin\left(u^2\right)\sin\left(u^1\right) \\ Z &= \cos\left(u^1\right) \end{aligned}$$

So, each point on the 2D plane u^1, u^2 gives us a vector $\vec{R}$ which lives in the X, Y, Z 3D space.
The u^1 curves are the lines of longitude running from north to south and the u^2 are the curves of latitude, running from west to east.
Also as a reminder, the tangent basis vectors on the sphere are given by the two partial derivatives of the position vector, where $\vec{e_1}$ are vectors tangent to the u^1 curves, and $\vec{e_2}$ are vectors tangent to the u^2 curves:

$$\frac{\partial \vec{R}}{\partial u^1} = \vec{e_1}$$
$$\frac{\partial \vec{R}}{\partial u^2} = \vec{e_2}$$

And now let's try to compute the covariant derivative of some tangent vector field $\vec{v}$ in the direction of the u^i coordinate.
Please note that we're considering the **vector field to be tangent to the surface**, and in most physics application, this is indeed the case for curved surfaces.

Allright, so remind yourself that the covariant derivative, we just said it's the ordinary derivative with the normal component subtracted:

$$\nabla_{\frac{\partial}{\partial u^i}} \vec{v} = \frac{\partial \vec{v}}{\partial u^i} - \left(\frac{\partial \vec{v}}{\partial u^i} \cdot \hat{n} \right) \hat{n}$$

To keep the notation lighter, let's call the normal part $\vec{N_i} \equiv \left(\frac{\partial \vec{v}}{\partial u^i} \cdot \hat{n} \right) \hat{n}$. Being the vector field composed by vectors always tangent to the surface, we can expand them as a linear combination of the basis vectors:

$$\begin{aligned}
\nabla_{\frac{\partial}{\partial u^i}} \vec{v} &= \frac{\partial \vec{v}}{\partial u^i} - \vec{N_i} \\
&= \frac{\partial}{\partial u^i} \left[v^j \vec{e_j} \right] - \vec{N_i} \\
&= \left[\frac{\partial v^j}{\partial u^i} \vec{e_j} + v^j \frac{\partial \vec{e_j}}{\partial u^i} \right] - \vec{N_i}
\end{aligned}$$

As you can remember, we can write down the second term in the squared brackets, the derivative of the basis vectors, as a linear combination of the Christoffel symbols and Second Fundamental form:

$$\begin{aligned}
\nabla_{\frac{\partial}{\partial u^i}} \vec{v} &= \left[\frac{\partial v^j}{\partial u^i} \vec{e_j} + v^j \frac{\partial \vec{e_j}}{\partial u^i} \right] - \vec{N_i} \\
&= \left[\frac{\partial v^j}{\partial u^i} \vec{e_j} + v^j \Gamma^k_{ij} \vec{e_k} \right] + v^j L_{ij} \hat{n} - \vec{N_i}
\end{aligned}$$

We can now factor out the $\vec{e_k}$ changing the summation index of the first term, which is just a dummy one:

$$\nabla_{\frac{\partial}{\partial u^i}} \vec{v} = \left[\frac{\partial v^k}{\partial u^i} + v^j \Gamma^k_{ij} \right] \vec{e_k} + v^j L_{ij} \hat{n} - \vec{N_i}$$

The term in square brackets is a combination of tangent vectors, so the only normal piece of $\frac{\partial \vec{v}}{\partial u^i}$ is $v^j L_{ij} \hat{n}$, which means $\vec{N_i} = v^j L_{ij} \hat{n}$. The last two terms cancel out, and what remains is just the tangential component:

$$\nabla_{\frac{\partial}{\partial u^i}} \vec{v} = \left[\frac{\partial v^k}{\partial u^i} + v^j \Gamma^k_{ij} \right] \vec{e_k}$$

which is our final formula for the covariant derivative.

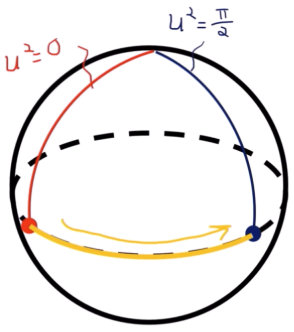
Before, in the geodesic part, we already derived how to compute the Christoffel symbols, so now we're ready to go proceed and actually calculate the covariant derivative in this case. We also already calculated the symbols for the sphere of radius one, and the only non-zero ones are:

$$\Gamma^1_{22} = -\cos(u)\sin(u)$$
$$\Gamma^2_{12} = \Gamma^2_{21} = \frac{\cos(u)}{\sin(u)}$$

Our summations run over i, j and k for the covariant derivative, but considering only the non-zero terms, we're left with the following two formulas for $i = 1$ (u^1), and $i = 2$ (u^2):

$$\nabla_{\vec{e_1}} \vec{v} = \frac{\partial v^1}{\partial u^1} \vec{e_1} + \left[\frac{\partial v^2}{\partial u^1} + v^2 \cot\left(u^1\right) \right] \vec{e_2}$$
$$\nabla_{\vec{e_2}} \vec{v} = \left[\frac{\partial v^1}{\partial u^2} - \frac{1}{2} v^2 \sin\left(2u^1\right) \right] \vec{e_1} + \left[\frac{\partial v^2}{\partial u^2} + v^1 \cot\left(u^1\right) \right] \vec{e_2}$$

A practical example would be considering curve running from $u^2 = 0$ to $u^2 = \pi/2$, with constant $u^1 = \pi/2$. Practically a curve where when the parameter increases, we run along the equator:



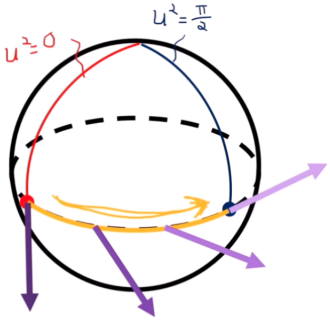
So the parametrization is:

$$\lambda \to \left(u^1 = \pi/2, u^2 = \lambda\right) \tag{63}$$

If we now consider a hypothetical vector field that is equal to $\vec{e_1}$ when $u^2 = 0$, and equal to $\vec{e_2}$ when $u^2 = \pi/2$:

$$\vec{v} = \cos\left(u^2\right)\vec{e_1} + \sin\left(u^2\right)\vec{e_2} \tag{64}$$

So this is a vector field that's definitely changing while going along the equator curve above:



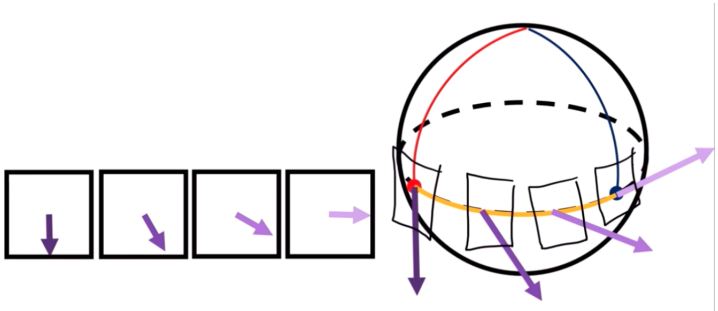
Intuitively then, we should expect its covariant derivative in the direction of the curve, to be different than zero, as this is clearly not a parallel transport:

$$\begin{aligned}\nabla_{\frac{\partial}{\partial\lambda}}\vec{v} = \nabla_{\frac{\partial}{\partial u^2}}\vec{v} = \nabla_{\vec{e_2}}\vec{v} \\ = \left[\frac{\partial v^1}{\partial u^2} - \frac{1}{2}v^2\sin\left(2u^1\right)\right]\vec{e_1} + \left[\frac{\partial v^2}{\partial u^2} + v^1\cot\left(u^1\right)\right]\vec{e_2}\end{aligned}$$

Since u^1 is constant and equal to $\pi/2$, the term with sin and cot go to zero, then:

$$\begin{aligned}\nabla_{\vec{e_2}}\vec{v} = \left[\frac{\partial v^1}{\partial u^2}\right]\vec{e_1} + \left[\frac{\partial v^2}{\partial u^2}\right]\vec{e_2} \\ = -\sin\left(u^2\right)\vec{e_1} + \cos\left(u^2\right)\vec{e_2}\end{aligned}$$

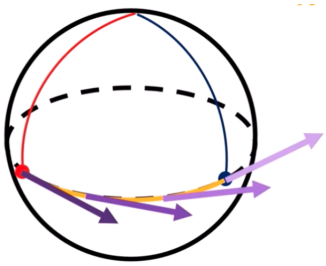
Which is indeed confirming that the covariant derivative is non-zero, and **it makes completely sense if you consider the tangent planes for each of the vectors moving along the curve; They're sort of spinning around in the tangent planes, so its rate of change is not completely normal to the surface, there's a tangential part that exists in the tangent plane**



If you instead, consider another vector field, simply described by:

$$\vec{v} = \vec{e_2} \tag{65}$$

along the same curve



And you try to compute its covariant derivative, you can easily see that all the terms go to zero. There's no vector change in the tangent planes, the whole rate of change happens along the normal direction with respect to the surface: we are parallel transporting this vector along the curve.

SO what we've done so far was, starting from a given vector field, we computed the covariant derivative and from its result, we checked if the vector field involves the parallel transport of a vector. What we'd like to do now is go in the opposite direction, so starting from a single vector, and given a curve, we will demand the parallel transport condition along a curve, and get the resulting vector field.

Let's start with the simple initial vector $\vec{v_0}$, on the equator curve described above, pointing along $\vec{e_1}$ at the origin point of the curve. For this, we demand the parallel transport condition of the covariant derivative along the curve to be zero:

$$\lambda \to \left(u^1 = \pi/2, u^2 = \lambda\right) \tag{66}$$

$$\vec{v_0} = \vec{e_1} \tag{67}$$

$$\nabla_{\vec{e_2}} \vec{v} = \vec{0} \tag{68}$$

We then start with the formula we used above, as before, the sin and cot terms go to zero, due to u^1 being constant and equal to $\pi/2$:

$$\nabla_{\vec{e_2}} \vec{v} = \left[\frac{\partial v^1}{\partial u^2}\right] \vec{e_1} + \left[\frac{\partial v^2}{\partial u^2}\right] \vec{e_2} = \vec{0}$$

What this requires, is that the derivatives of both vector field components are equal to zero, which means the components have to be some constants:

$$\frac{\partial v^1}{\partial u^2} = 0$$
$$\frac{\partial v^2}{\partial u^2} = 0$$

$$v^1(\lambda) = c_1$$
$$v^2(\lambda) = c_2$$

Remember that we need to respect the initial condition right? At $\lambda = 0$ we had that $v^1 = 1$ and $v^2 = 0$, which simply means that the vector components have to always be constant and equal to:

$$v^1(\lambda) = c_1 = 1$$
$$v^2(\lambda) = c_2 = 0$$

$$\vec{v}(\lambda) = \vec{e_1} \tag{69}$$

This is the vector field resulting by requiring the parallel transport of the initial vector given.

9.3 Covariant derivative in curved-space - intrinsic

Allright and here we are with another definition of the covariant derivative, this time using intrinsic geometry.
You may wonder why we need this in the first place, if the extrinsic defintion gets the job done; Well the reason is that there are cases, like General Relativity, where we cannot use any extrinsic world embedding our 2D surface, simply because it does not exist such thing. In GR, spacetime is a 4 dimentional space, and when Einstein explained his theory, it did not think about this space being inside a higher dimentional one, where we could leverage the use of an origin or position vectors.
The universe itself, the spacetime, is intrinsically curved with no outside space existing at all.

So, when working using intrinsic geometry, we can forget about a higher dimentional XYZ coordinates we can use, we can forget about position vectors, and global origin from where to define this. Remember that intrinsic can also be seen as a "bug's" view of the surface, where basis vectors are defined as partial derivative operators, and not derivative of the position vector like in extrinsic geometry. They live in the Tangent Vector Space T_pM at some point p.

So, let's try to compute the formula for the covariant derivative for intrinsically curved spaces. Before, we said that the covariant derivative of a vector field, is the ordinary derivative, with the normal component of the rate of change subtracted, but since we're dealing with intrinsic geometry now, there's no such thing as normal vectors, as this would imply there's an outside space we can point to, and this is not the case now.

So we ignore the normal part of the formula, and we can try writing down the covariant derivative, with the Christoffel symbols used to represent the basis vectors derivative coefficients, and then just re-arranging the dummy index to factor out the basis vectors:

$$
\begin{aligned}
\nabla_{\frac{\partial}{\partial u^i}} \vec{v} &= \frac{\partial \vec{v}}{\partial u^i} \\
&= \frac{\partial}{\partial u^i} \left(v^j \vec{e_j} \right) \\
&= \frac{\partial v^j}{\partial u^i} \vec{e_j} + v^j \frac{\partial \vec{e_j}}{\partial u^i} \\
&= \left(\frac{\partial v^k}{\partial u^i} + v^j \Gamma^k_{ij} \right) \vec{e_k}
\end{aligned}
$$

Now, the problem ends up being the Christoffel symbols, because before, when we derived the expression for these, we ended up with this formula, which requires the dot product of the basis vectors with the derivative of the basis vectors:

$$
\Gamma^m_{ij} = \left(\frac{\partial \vec{e_j}}{\partial u^i} \cdot \vec{e_l} \right) \mathfrak{g}^{lm} \tag{70}
$$

Since we don't have now, the XYZ extrinsic basis, we can't easily compute these dot products, so we need a new strategy to compute the Christoffel symbols.
There are three important facts to recognize before we proceed:

- The metric tensor components are the dot products of the basis vectors $g_{ij} = \vec{e_i} \cdot \vec{e_j}$
- The order of the dot product does not matter, hence the metric tensor components are symmetric in the i and j indexes $g_{ij} = g_{ji}$
- The Christoffel symbols are symmetric in their lower indexes, so $\Gamma^k_{ij} = \Gamma^k_{ji}$

Now, we start by taking the derivative of the metric tensor in the direction of the u^k coordinate, where we can leverage the metric tensor definition, and an important property called **metric compatibility** (more on this in the next section, we take it for granted now), which allows us to use the derivative product rule applied to dot products:

$$
\begin{aligned}
\frac{\partial}{\partial u^k} g_{ij} &= \frac{\partial}{\partial u^k} \left(\vec{e_i} \cdot \vec{e_j} \right) \\
&= \frac{\partial \vec{e_i}}{\partial u^k} \cdot \vec{e_j} + \vec{e_i} \cdot \frac{\partial \vec{e_j}}{\partial u^k} \\
&= \left(\Gamma^l_{ik} \vec{e_l} \right) \cdot \vec{e_j} + \vec{e_i} \cdot \left(\Gamma^l_{jk} \vec{e_l} \right) \\
&= \Gamma^l_{ik} \left(\vec{e_l} \cdot \vec{e_j} \right) + \Gamma^l_{jk} \left(\vec{e_i} \cdot \vec{e_l} \right) \\
&= \Gamma^l_{ik} g_{lj} + \Gamma^l_{jk} g_{il}
\end{aligned}
$$

We're free to swap the lower indexes of the metric tensor, so we'll do so for the first term and we obtain:

$$
\frac{\partial g_{ij}}{\partial u^k} = \Gamma^l_{ik} g_{jl} + \Gamma^l_{jk} g_{il} \tag{71}
$$

Now, note this down first, and then we're going to take other derivatives of the metric tensor, twice more, but with different indexes:

$$\frac{\partial g_{ki}}{\partial u^j} = \Gamma^l_{kj} g_{il} + \Gamma^l_{ij} g_{kl}$$
$$\frac{\partial g_{jk}}{\partial u^i} = \Gamma^l_{ji} g_{kl} + \Gamma^l_{ki} g_{jl}$$

Let's watch carefully the three together, and I will color code the terms here:

$$\frac{\partial g_{ij}}{\partial u^k} = \Gamma^l_{ik} g_{jl} + \Gamma^l_{jk} g_{il}$$
$$\frac{\partial g_{ki}}{\partial u^j} = \Gamma^l_{kj} g_{il} + \Gamma^l_{ij} g_{kl}$$
$$\frac{\partial g_{jk}}{\partial u^i} = \Gamma^l_{ji} g_{kl} + \Gamma^l_{ki} g_{jl}$$

The terms with the same color, are actually the same, so we can do something here, let's sum them in this way:

$$\frac{\partial g_{ij}}{\partial u^k} + \frac{\partial g_{ki}}{\partial u^j} - \frac{\partial g_{jk}}{\partial u^i} = \Gamma^l_{jk} g_{il} + \Gamma^l_{kj} g_{il}$$
$$\frac{\partial g_{ij}}{\partial u^k} + \frac{\partial g_{ki}}{\partial u^j} - \frac{\partial g_{jk}}{\partial u^i} = 2\Gamma^l_{jk} g_{li}$$

We can now sum on both sides, with the inverse metric tensor:

$$\mathfrak{g}^{im}\left(\frac{\partial g_{ij}}{\partial u^k} + \frac{\partial g_{ki}}{\partial u^j} - \frac{\partial g_{jk}}{\partial u^i}\right) = 2\Gamma^l_{jk} g_{li} \mathfrak{g}^{im}$$

Using the fact that the these metric tensors are inverse of each other, we finally obtain our new formula for the Christoffel symbols, only dependent on the metric tensor and its inverse:

$$\Gamma^m_{jk} = \frac{1}{2}\mathfrak{g}^{im}\left(\frac{\partial g_{ij}}{\partial u^k} + \frac{\partial g_{ki}}{\partial u^j} - \frac{\partial g_{jk}}{\partial u^i}\right) \tag{72}$$

One last topic on this section, related to Geodesics. We previously saw that geodesics define the most straight path in a curve surface/space. We can define this idea of walking forward along a geodesic, using the covariant derivative.
This idea of walking forward on a curved space/surface is captured, mathematically, by the idea of parallel transporting a vector along itself. **When we say that a vector is parellel transported along itself, what we mean is that the vector we're taking the covariant derivative of, and the direction that we choose, are the same.**

So when we say that the covariant derivative of a vector in the direction of itself is equal to zero, the resulting curve we get is a geodesic curve.

If our curve is a function of the parameter λ, then the tangent vectors along the curve would be given by the derivative operators $\frac{d}{d\lambda}$. A geodesic curve is described by the covariant derivative $\nabla_{\frac{d}{d\lambda}} \frac{d}{d\lambda} = \vec{0}$.

Let's try this out, first expanding using the multi-variable chain-rule, and then the derivative product rule:

$$\begin{aligned}
\nabla_{\frac{d}{d\lambda}} \frac{d}{d\lambda} &= \frac{d}{d\lambda}\left(\frac{d}{d\lambda}\right) \\
\vec{0} &= \frac{d}{d\lambda}\left(\frac{du^j}{d\lambda}\frac{\partial}{\partial u^j}\right) \\
&= \frac{d}{d\lambda}\left(\frac{du^j}{d\lambda}\right)\frac{\partial}{\partial u^j} + \frac{du^j}{d\lambda}\frac{d}{d\lambda}\left(\frac{\partial}{\partial u^j}\right) \\
&= \frac{d^2u^j}{d\lambda^2}\frac{\partial}{\partial u^j} + \frac{du^j}{d\lambda}\frac{d}{d\lambda}\left(\frac{\partial}{\partial u^j}\right) \\
&= \frac{d^2u^j}{d\lambda^2}\frac{\partial}{\partial u^j} + \frac{du^j}{d\lambda}\left(\frac{du^i}{d\lambda}\frac{\partial}{\partial u^i}\right)\left(\frac{\partial}{\partial u^j}\right) \\
&= \frac{d^2u^j}{d\lambda^2}\frac{\partial}{\partial u^j} + \frac{du^i}{d\lambda}\frac{du^j}{d\lambda}\left(\frac{\partial^2}{\partial u^i \partial u^j}\right)
\end{aligned}$$

We can just re-write it now, using the basis vector notation and then use the Christoffel simbols to express the derivative of the basis vectors:

$$\begin{aligned}
\vec{0} &= \frac{d^2u^j}{d\lambda^2}\vec{e_j} + \frac{du^i}{d\lambda}\frac{du^j}{d\lambda}\left(\frac{\partial \vec{e_j}}{\partial u^i}\right) \\
&= \frac{d^2u^j}{d\lambda^2}\vec{e_j} + \frac{du^i}{d\lambda}\frac{du^j}{d\lambda}\Gamma^k_{ij}\vec{e_k}
\end{aligned}$$

We can just replace the first term's dummy index with k and factor the $\vec{e_k}$ basis vector out:

$$\vec{0} = \left(\frac{d^2u^k}{d\lambda^2} + \frac{du^i}{d\lambda}\frac{du^j}{d\lambda}\Gamma^k_{ij}\right)\vec{e_k}$$

This is, as you may remember, exactly the geodesic equation. It's just that now, with the covariant derivative notion, we can state that the **geodesic is the curve resulting from parallel transporting a vector along itself.**

9.4 Covariant derivative - abstract definition

Now, finally, after having defined the covariant derivative first in flat space, then in curved spaces extrinsically, and afterward intrinsically, we're ready to give the most abstract, and at the same time, most powerful, definition of covariant derivative.

To save up some notation heavyness, we will use interchangeably, the following representations of the basis vectors:

$$\vec{e_k} \equiv \frac{\partial}{\partial u^k} \equiv \partial_k \tag{73}$$

There are two main reasons why we want to study this abstract definition:

- We're going to find that there are actually "several" covariant derivatives, not just one; The one we care about is called the Levi-Civita connection, which comes from the Fundamental Theorem of Riemannian Geometry.
- It extends the covariant derivative to all tensor fields.

To help understanding how we arrived at the definition of the abstract covariant derivative, it helps to think about how another concept has been invented in the first place: the concept of vector spaces.

Let's start thinking about $\mathbb{R}^2$, the set of pairs of real numbers. The members of this set serve as an inspiration to the vector space idea. If we start checking some properties of this set, you can clearly see that it's possible to find a reasonable way of adding two members of this set to obtain another member of the set, also there's a reasonable way to scale the members by multiplying them by a scalar element, and obtain new members of the set.

Moreover, the order of addition does not matter, and we also find that there's a special zero element, that if we add to a member pair, gives back the same member pair. Each element also, had a negative version of it, that when added to the opposite member pair, gives the zero element.

On top of this, we have distribution properties for multiplication over addition:
For $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^2$:

- $\vec{v} + \vec{w} \in \mathbb{R}^2$
- $\vec{v} + \vec{w} = \vec{w} + \vec{v}$
- $\vec{u} + (\vec{v} + \vec{w}) = (\vec{u} + \vec{v}) + \vec{w}$
- $\exists \vec{0} \in \mathbb{R}^2, \vec{v} + \vec{0} = \vec{v}$
- $\exists (-\vec{v}) \in \mathbb{R}^2, \vec{v} + (-\vec{v}) = \vec{0}$

For $\vec{v} \in \mathbb{R}^2, a, b \in \mathbb{R}$:

- $a\vec{v} \in \mathbb{R}^2$
- $a(b\vec{v}) = (ab)\vec{v}$
- $1\vec{v} = \vec{v}$

For $\vec{v}, \vec{w} \in \mathbb{R}^2; a, b \in \mathbb{R}$:

- $a(\vec{v} + \vec{w}) = a\vec{v} + a\vec{w}$
- $(a + b)\vec{v} = a\vec{v} + b\vec{v}$

We can take all these properties and turn them into a definition, we say that a vector space is a set of vectors and a set of scalars, along with rules of addition and scaling, that follow the properties above.

So you see what the process was: we started with the properties of $\mathbb{R}^2$, and use them to define a vector space, which is a more abstract idea. Now, given this new definition, we can find other examples of other things in mathematics, that obey the same rules, like the set of N-tuples of real numbers, set of polynomials in one variable, the set of partial derivatives of coordinate variables, etc..

So, inventing the abstract definition of the covariant derivative, basically follows the same process. We start with the definition of the covariant derivative we're familiar with, look at some of its properties and then use those properties to define an abstract definition for it. From there then, we can define many other types of covariant derivatives as well.

So we start by looking at some properties of the covariant derivatives:

- $\nabla_{a\vec{w}+b\vec{t}}\vec{v} = a\nabla_{\vec{w}}\vec{v} + b\nabla_{\vec{t}}\vec{v}$
- $\nabla_{\vec{w}}(\vec{v} + \vec{u}) = \nabla_{\vec{w}}\vec{v} + \nabla_{\vec{w}}\vec{u}$
- $\nabla_{\vec{w}}(a\vec{v}) = (\nabla_{\vec{w}}a)\vec{v} + a(\nabla_{\vec{w}}\vec{v})$
- $\nabla_{\partial_i}(a) = \frac{\partial a}{\partial u^i}$

Note that in the first and third properties, a and b are allowed to be scalar *fields* (functions on the surface), not just constants. In particular, $\nabla_{a\vec{w}}\vec{v} = a\nabla_{\vec{w}}\vec{v}$ even when a varies from point to point: scaling the direction only scales the result.

So we're going to consider these 4 properties and take them as a definition of the abstract covariant derivative. A covariant derivative is an operator with two input slot, it takes a direction vector, and an input field, and the output will be the rate of change of that field along that direction, and this will be a new field.

Sometimes, this covariant derivative operator, is also called "Connection". This is because of the parallel transport condition. We saw that in order to parallel transport a vector field along a curve, its covariant derivative has to be zero, if you look at this in terms of tangent vector spaces where we start and we end along the curve, the parallel transport basically provides a connection between tangent spaces in a curved space.

If you start at a point p, with the tangent space defined by T_pM, and end up at the point q with tangent space T_qM, the coveriant derivative provides you a connection between these two tangent spaces in the curved space.

Now you remember how we defined the Christoffel symbols, well in this case we can simply write, in the most abstract way:

$$\nabla_{\vec{e_i}}\vec{e_j} = \Gamma^k_{ij}\vec{e_k} \tag{74}$$

Sometimes these are also called connection coefficients, and it turns out that, given the information we defined so far, we don't have enough information on how to solve for a unique set of these symbols, that's why we stated at the beginning of this section, that there are many possible covariant derivatives, it's because there are many possible ways to define these connection coefficients.

Though, there's a way we can get a unique solution for the Christoffel symbols, and that's by introducing two new properties called:

- The torsion-free property;
- The metric compatibility property.

The torsion-free property says that, the covariant derivative of the basis vector $\vec{e_j}$ along the direction $\vec{e_i}$, is the same as the covariant derivative of the basis vector $\vec{e_i}$ along the direction $\vec{e_j}$:

$$\nabla_{\vec{e_i}}\vec{e_j} = \nabla_{\vec{e_j}}\vec{e_i} \tag{75}$$

This simple form holds for a *coordinate* basis ($\vec{e_i} = \partial_i$), where partial derivatives commute ($\partial_i\partial_j = \partial_j\partial_i$). For general vector fields the torsion-free condition reads $\nabla_{\vec{u}}\vec{w} - \nabla_{\vec{w}}\vec{u} = [\vec{u}, \vec{w}]$, where $[\vec{u}, \vec{w}]$ is the Lie bracket, which vanishes for coordinate basis vectors.

This is what let us write, that the Christoffel symbols are symmetric on their lower indexed:

$$\nabla_{\vec{e_i}}\vec{e_j} = \Gamma^k_{ij}\vec{e_k}$$
$$\nabla_{\vec{e_j}}\vec{e_i} = \Gamma^k_{ji}\vec{e_k}$$

$$\boxed{\Gamma^k_{ij} = \Gamma^k_{ji}} \tag{76}$$

The metric compatibility property is a sort of a special version of the derivation product rule, that's applied to the dot product:

$$\nabla_{\vec{w}}(\vec{v}\cdot\vec{u}) = (\nabla_{\vec{w}}\vec{v})\cdot\vec{u} + \vec{v}\cdot(\nabla_{\vec{w}}\vec{u}) \tag{77}$$

This might be a weird rule, but in plain english, what this is saying is that **when we parallel transport two vectors, their dot product will stay the same.**

If you take these two vectors $\vec{v}$ and $\vec{u}$ and you parallel transport them, the condition is that their covariant derivative is $\vec{0}$, so if you plug this result into the metric compatibility property, you find that the covariant derivative of their dot product is zero ($\nabla_{\vec{w}}(\vec{v} \cdot \vec{u}) = 0$), which simply means that the dot product stays constant. Which eventually means that the length of a parallel transported vector stays constant, and in case of considering two vectors, if parallel transported, their angle in between, stays constant.

One consequence of the metric compatibility property can be obtained if we replace the vectors with the basis vectors:

$$\nabla_{\vec{e_k}}(\vec{e_i} \cdot \vec{e_j}) = (\nabla_{\vec{e_k}} \vec{e_i}) \cdot \vec{e_j} + \vec{e_i} \cdot (\nabla_{\vec{e_k}} \vec{e_j}) \tag{78}$$

We can replace the covariant derivatives with the Christoffel symbols:

$$\nabla_{\vec{e_k}}(\vec{e_i} \cdot \vec{e_j}) = (\Gamma^l_{ik} \vec{e_l}) \cdot \vec{e_j} + \vec{e_i} \cdot (\Gamma^l_{jk} \vec{e_l}) \tag{79}$$

The dot product is a scalar, so we can simply replace the covariant derivative with a partial derivative:

$$\partial_k(\vec{e_i} \cdot \vec{e_j}) = (\Gamma^l_{ik} \vec{e_l}) \cdot \vec{e_j} + \vec{e_i} \cdot (\Gamma^l_{jk} \vec{e_l}) \tag{80}$$

Now, bringing the Christoffel symbols in front, will show the dot products, that we can simply re-write as metric tensor components:

$$\begin{aligned} \partial_k(\vec{e_i} \cdot \vec{e_j}) &= \Gamma^l_{ik}(\vec{e_l} \cdot \vec{e_j}) + \Gamma^l_{jk}(\vec{e_i} \cdot \vec{e_l}) \\ \partial_k(g_{ij}) &= \Gamma^l_{ik}(g_{lj}) + \Gamma^l_{jk}(g_{il}) \end{aligned}$$

Using the same, and switching around indexes, we can write down the following 3 equations then, and using the torsion-free property, we can color code the Christoffel symbols that are the same, due to the symmetry of their lower indexes:

$$\begin{aligned} \partial_k g_{ij} &= \Gamma^l_{ik} g_{jl} + \Gamma^l_{jk} g_{il} \\ \partial_j g_{ki} &= \Gamma^l_{kj} g_{il} + \Gamma^l_{ij} g_{kl} \\ \partial_i g_{jk} &= \Gamma^l_{ji} g_{kl} + \Gamma^l_{ki} g_{jl} \end{aligned}$$

If we try to add them in this way

$$\partial_k g_{ij} + \partial_j g_{ki} - \partial_i g_{jk} \tag{81}$$

We clearly see that the blue and teal terms cancel, and we're left with:

$$\partial_k g_{ij} + \partial_j g_{ki} - \partial_i g_{jk} = 2\Gamma^l_{jk} g_{il} \tag{82}$$

And then summing on both sides for the inverse metric tensor, we obtain a formula to calculate the Christoffel symbols:

$$\Gamma^m_{jk} = \frac{1}{2}\mathfrak{g}^{\mathfrak{im}}\left(\partial_k g_{ij} + \partial_j g_{ki} - \partial_i g_{jk}\right) \tag{83}$$

So using both properties, we're able to solve for the Christoffel symbols, which means we found one particular version of a covariant derivative, and this particular covariant derivative is called the Levi-Civita connection.

And this is basically the result stated in the **Fundamental Theorem of Riemannian Geometry** (where a Riemannian Manifold is basically a curved space with a metric): there's a unique connection (covariant derivative) that is:

- torsion-free
- has metric compatibility

And this connections is called the Levi-Civita connection, and the connection coefficients (Christoffel symbols) are given by the formula we found above.

As stated before, there are multiple other covariant derivatives apart from this, but only the Levi-Civita one is both torsion-free and metric compatible, and it's the one we're mainly interested in.

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